CSE472 (Machine Learning Sessional)

Assignment 3: Function Approximation with Neural Network and Backpropagation

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#1905108

HOW TO RUN:

Ensure the Jupyter Notebook ‘1905108.ipynb’ and Datasets Are in the Same Directory. Install pandas, numpy, scikit-learn, matplotlib, seaborn, pickle, tqdm, torchvision if not already installed. Then run the notebook ‘1905108.ipynb’.

The implementation of Dense layer, Normalization, Activation, Regularization, Optimization and Regression are in ‘Neural Net’ section.

The different model architecture can be found in ‘Different models’ section.

For different model architectures and various hyperparameter settings, 27 different models have been trained. Their training and validation accuracies are shown in the "27 Different Models" section.

In the ‘Final Test’ section best models performance has been shown.

PERFORMANCE EVALUATION (3 Different models):

Model 1

#layer 1

Dense(784, 392)

BatchNormalization()

Relu()

Dropout()

#layer 2

Dense(392, 196)

BatchNormalization()

Relu()

Dropout()

#layer 3

Dense(196, 98)

BatchNormalization()

Relu()

Dropout()

#layer 4

Dense(98,10)

BatchNormalization()

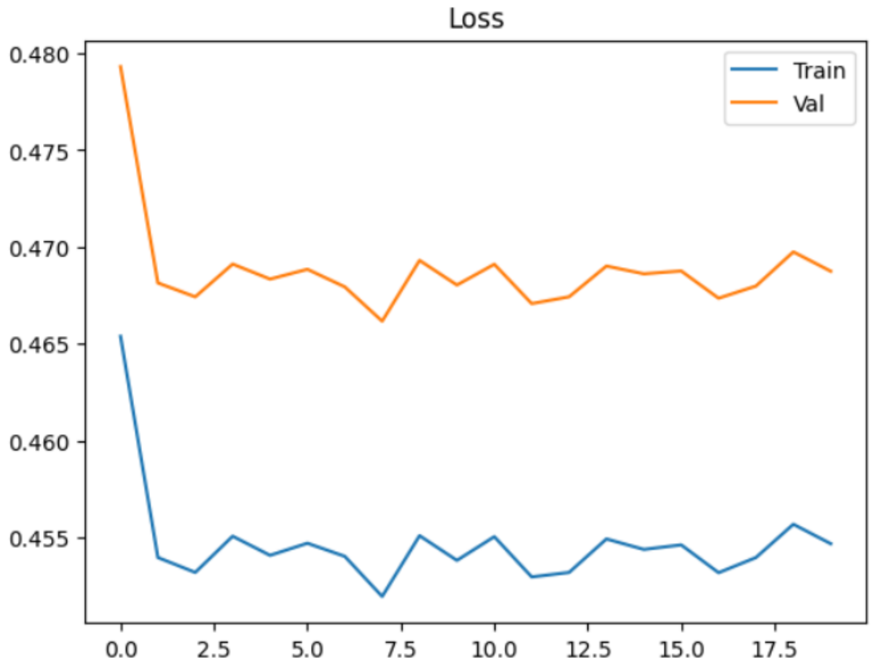
Softmax()

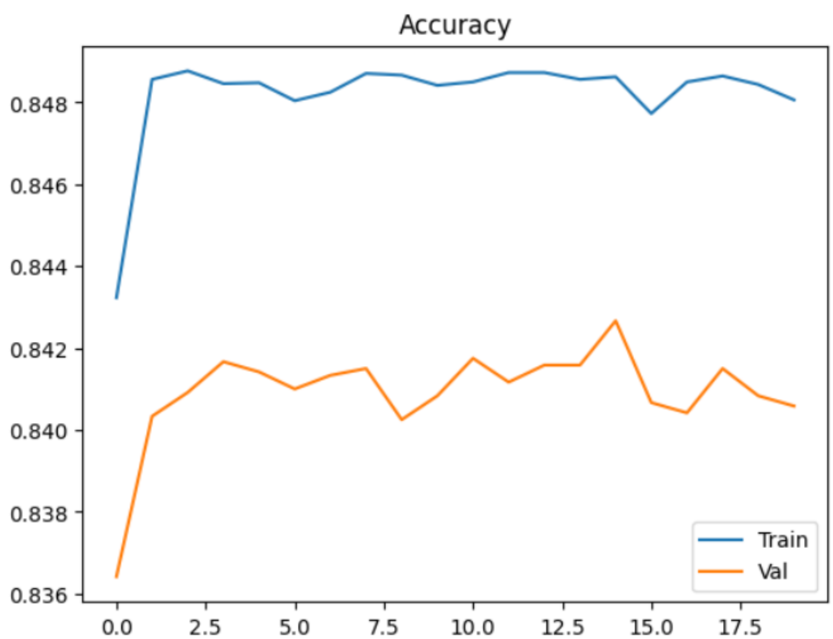
Sequential batch train:

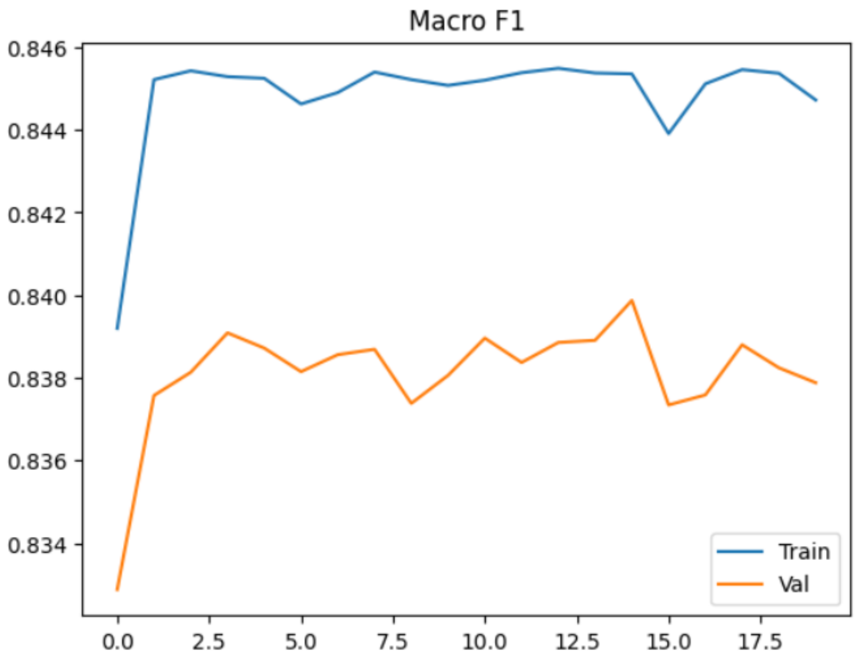
```
Model 1. 1
alpha:  0.01 decay: 0.99 dropout: 0.5 optimizer: ADAM  epoch: 20
```

Performance

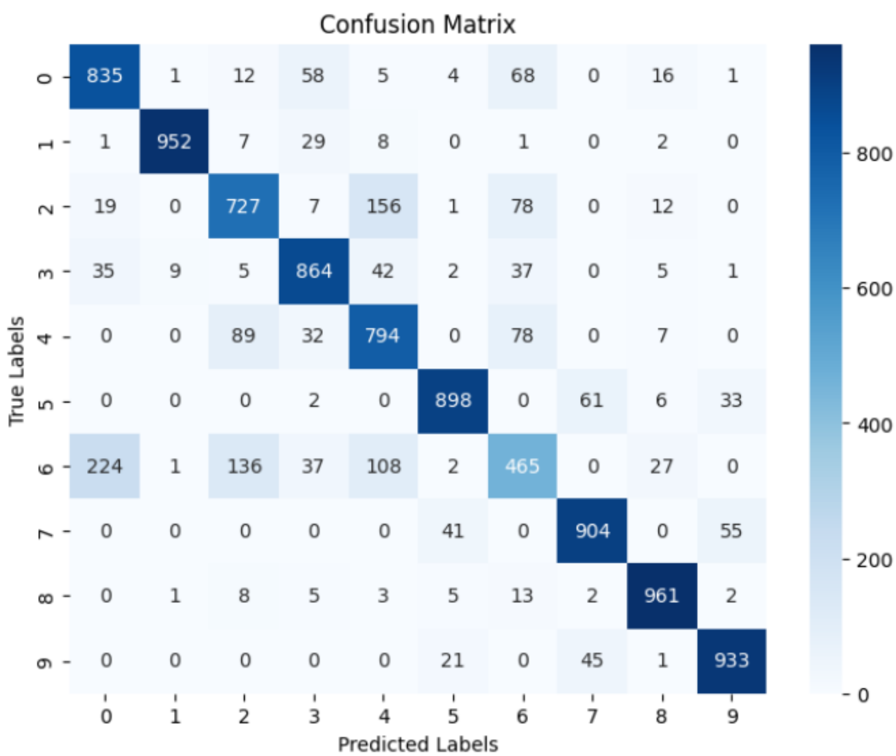
	Accuracy	Loss	Macro-F1-score
Train	0.8480625	0.454687	0.844720
Validation	0.840583	0.468747	0.837879
Test	0.8333	0.489168	0.830025


Loss
Train
Val
0.480
0.475
0.470
0.465
0.460
0.455
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Accuracy
Train
Val
0.848
0.846
0.844
0.842
0.840
0.838
0.836
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5



Macro F1
0.846
0.844
0.842
0.840
0.838
0.836
0.834
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Train
Val



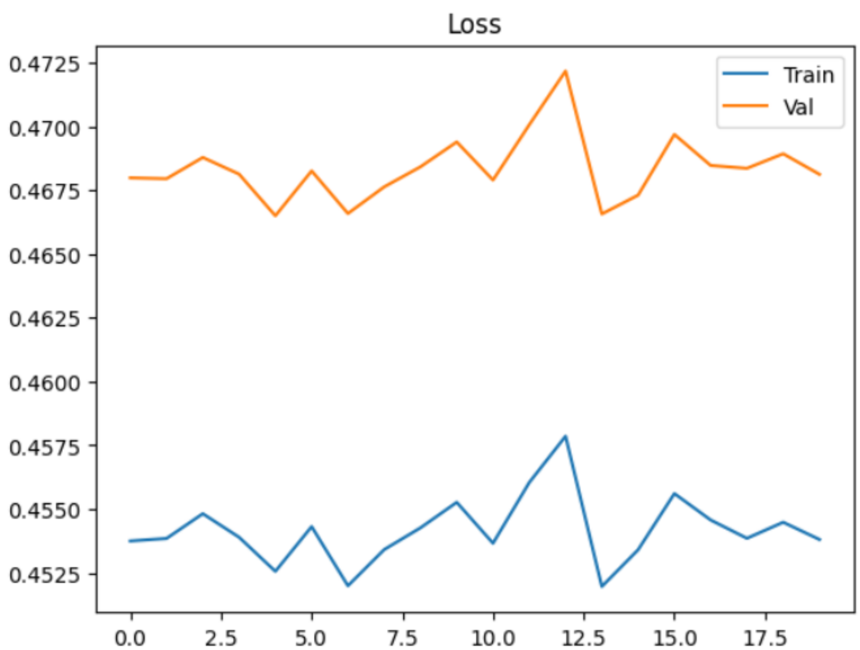
Confusion Matrix
True Labels
Predicted Labels
0 1 2 3 4 5 6 7 8 9
0: 835 1 12 58 5 4 68 0 16 1
1: 1 952 7 29 8 0 1 0 2 0
2: 19 0 727 7 156 1 78 0 12 0
3: 35 9 5 864 42 2 37 0 5 1
4: 0 0 89 32 794 0 78 0 7 0
5: 0 0 0 2 0 898 0 61 6 33
6: 224 1 136 37 108 2 465 0 27 0
7: 0 0 0 0 0 41 0 904 0 55
8: 0 1 8 5 3 5 13 2 961 2
9: 0 0 0 0 0 21 0 45 1 933
800
600
400
200
0

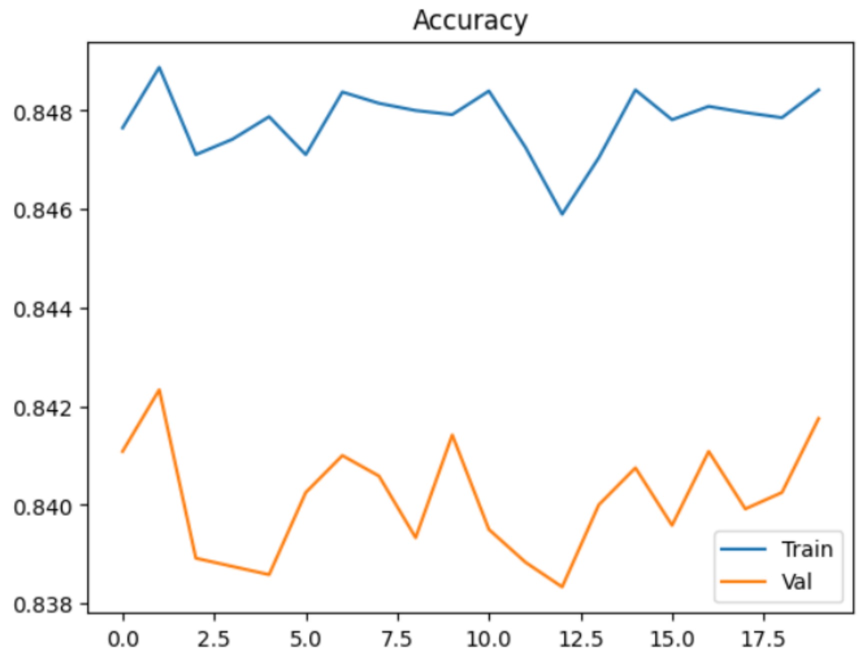
Random batch train:

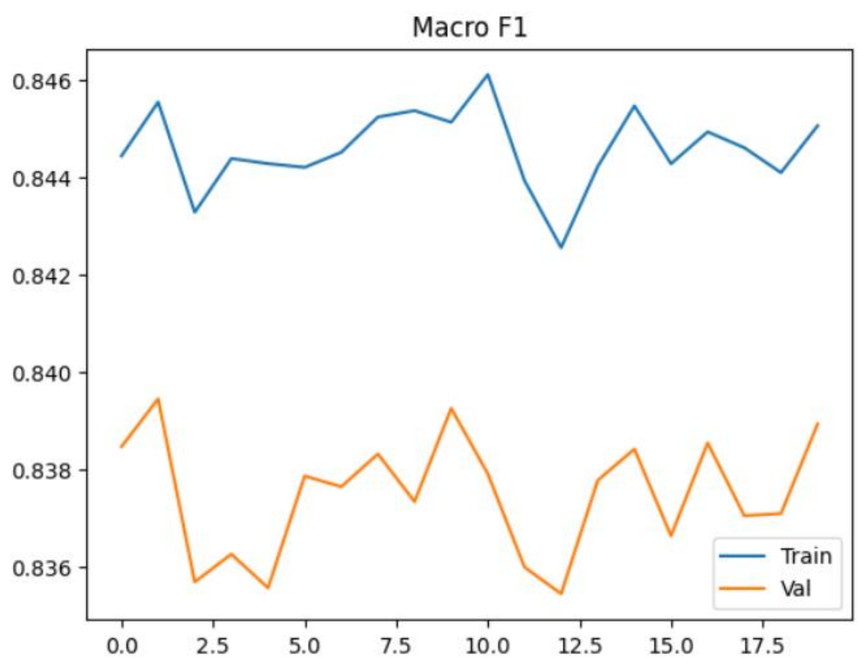
```
Model 1. 1
alpha:  0.01 decay: 0.99 dropout: 0.5 optimizer: ADAM  epoch: 20
```

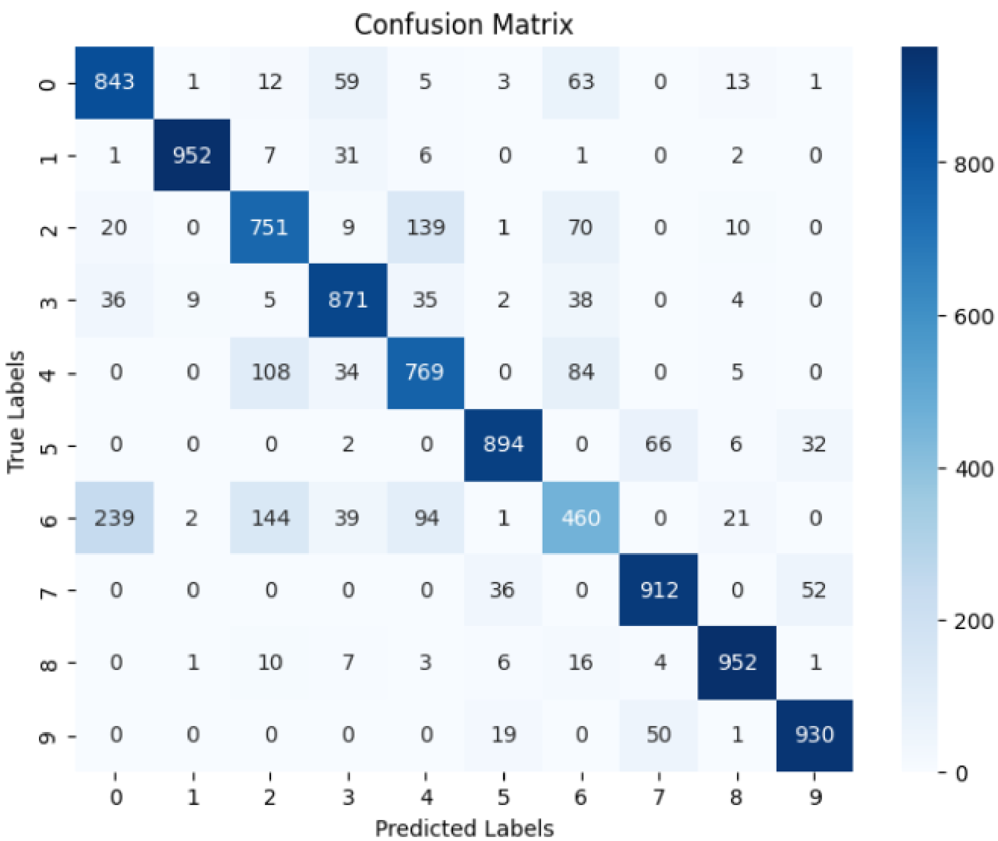
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.848416	0.453807	0.845058
Validation	0.84175	0.468124	0.838925
Test	0.8334	0.488021	0.830147




Accuracy
0.848
0.846
0.844
0.842
0.840
0.838
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Train
Val


Macro F1
0.846
0.844
0.842
0.840
0.838
0.836
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Train
Val

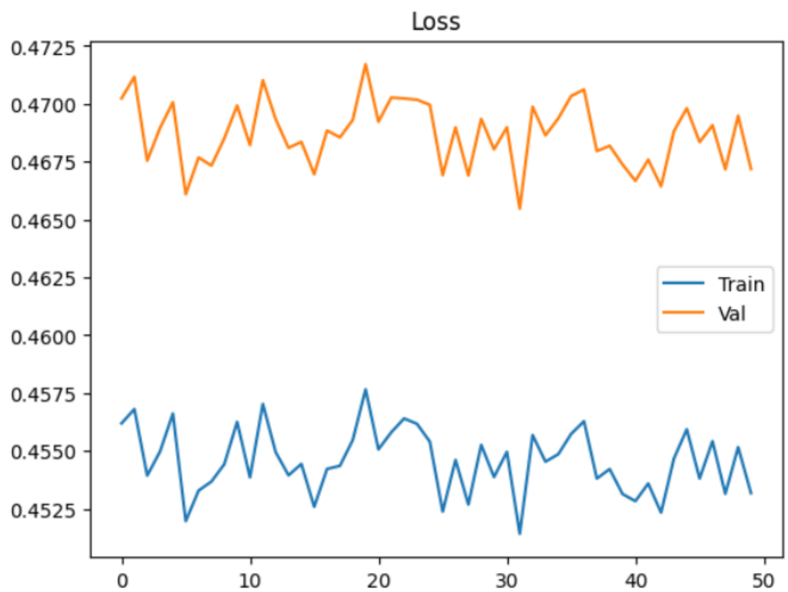


Random2 batch train:

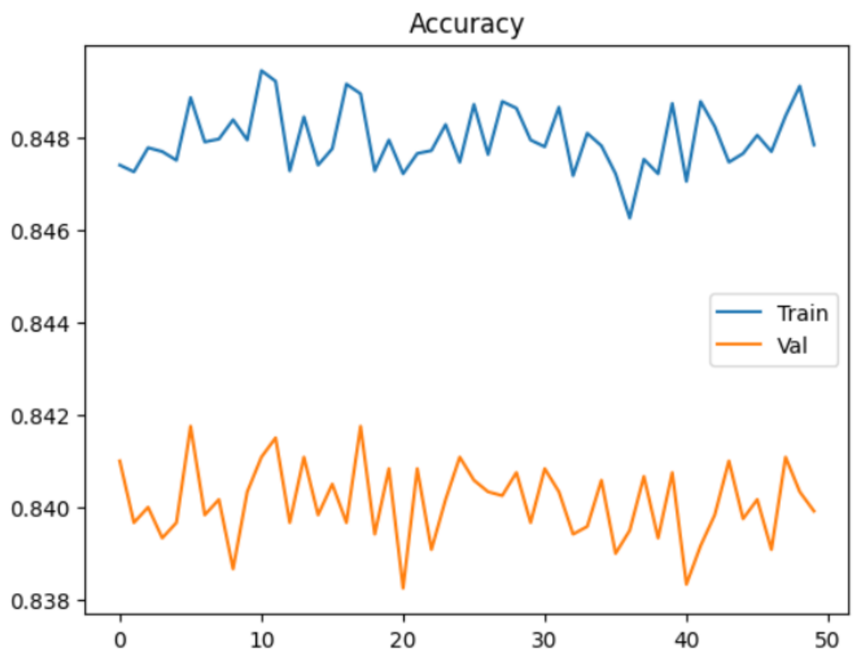
```
Model 1. 1
alpha:  0.01 decay: 0.99 dropout: 0.5 optimizer: ADAM  epoch: 50
```

Performance

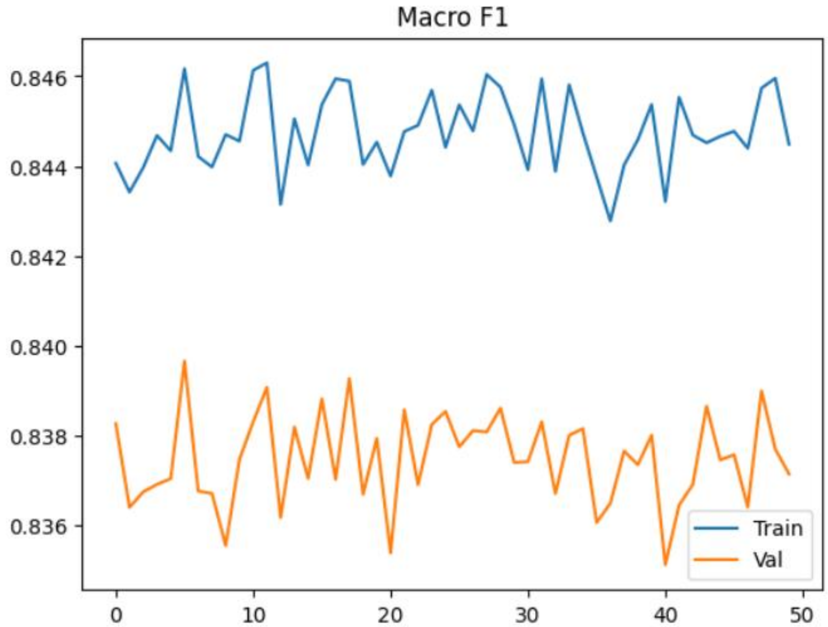
	Accuracy	Loss	Macro-F1-score
Train	0.847833	0.453180	0.844486
Validation	0.839916	0.467186	0.837136
Test	0.8342	0.487132	0.830916

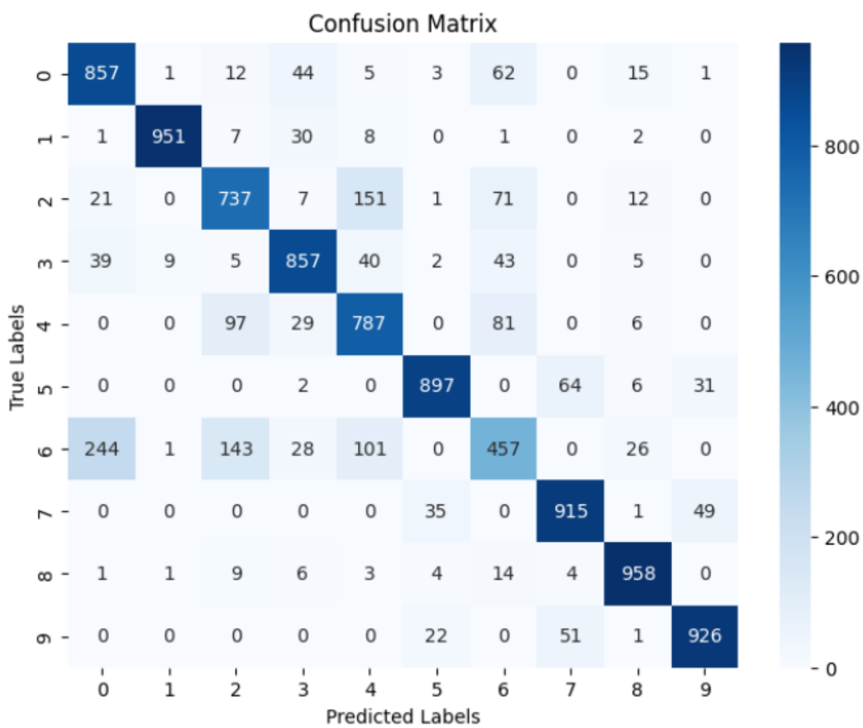


Loss
0.4725
0.4700
0.4675
0.4650
0.4625
0.4600
0.4575
0.4550
0.4525
0
10
20
30
40
50
Train
Val



Accuracy
0.848
0.846
0.844
0.842
0.840
0.838
0
10
20
30
40
50
Train
Val


Macro F1
0.846
0.844
0.842
0.840
0.838
0.836
0
10
20
30
40
50
Train
Val

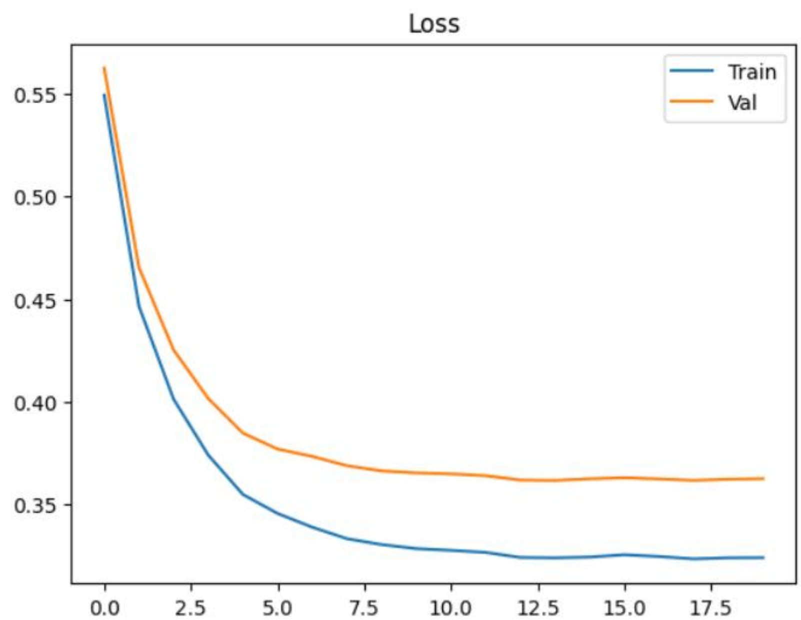

Confusion Matrix
True Labels
Predicted Labels
0 1 2 3 4 5 6 7 8 9
857 1 12 44 5 3 62 0 15 1
1 951 7 30 8 0 1 0 2 0
21 0 737 7 151 1 71 0 12 0
39 9 5 857 40 2 43 0 5 0
0 0 97 29 787 0 81 0 6 0
0 0 0 2 0 897 0 64 6 31
244 1 143 28 101 0 457 0 26 0
0 0 0 0 0 35 0 915 1 49
1 1 9 6 3 4 14 4 958 0
0 0 0 0 0 22 0 51 1 926
800
600
400
200
0

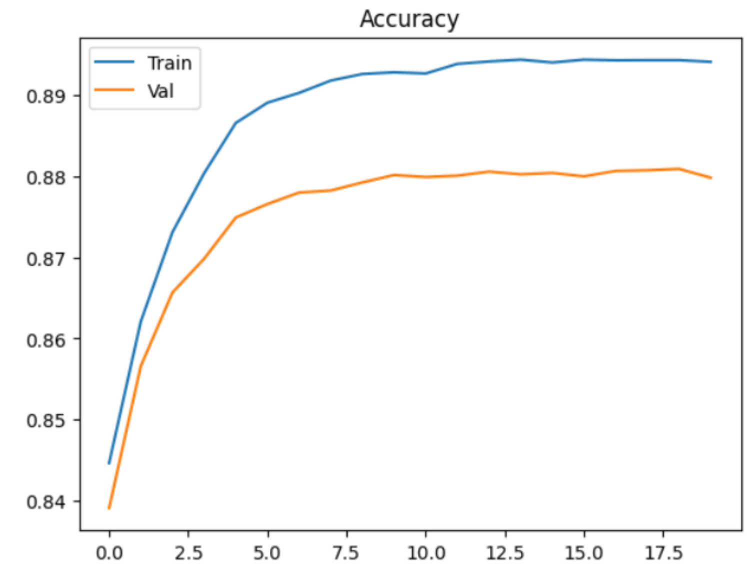
Sequential batch train:

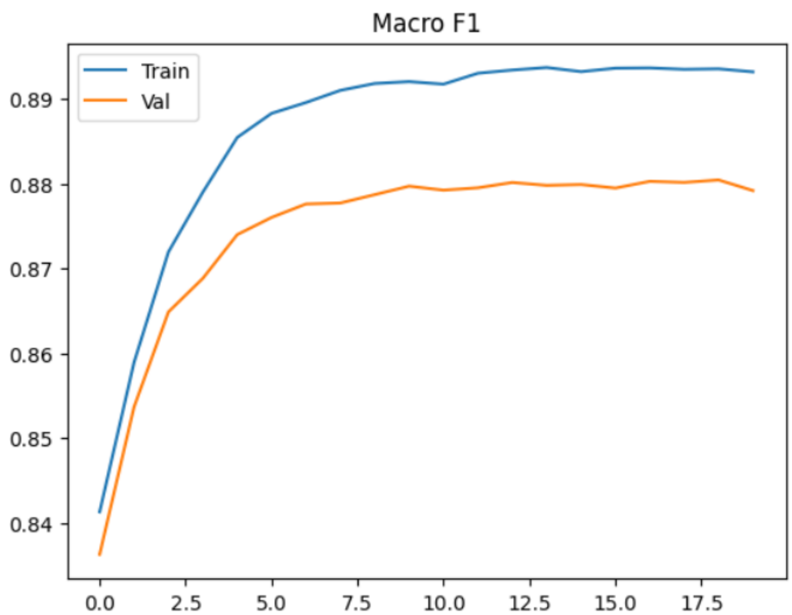
```
Model 1. 2
alpha:  0.001 decay: 0.999 dropout: 0.5 optimizer: ADAM  epoch: 20
```

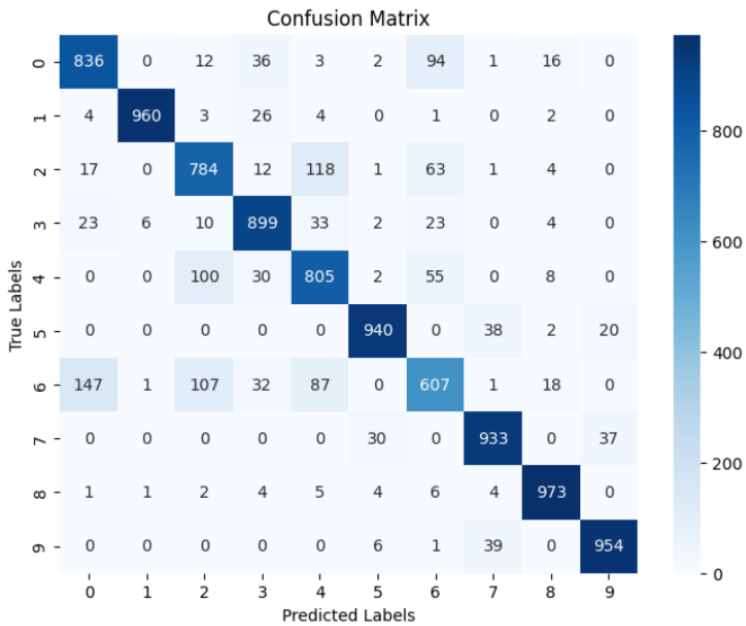
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.894125	0.324163	0.893171
Validation	0.879833	0.362587	0.879170
Test	0.8691	0.391021	0.867865




Accuracy
Train
Val
0.89
0.88
0.87
0.86
0.85
0.84
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Macro F1
Train
Val
0.89
0.88
0.87
0.86
0.85
0.84
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

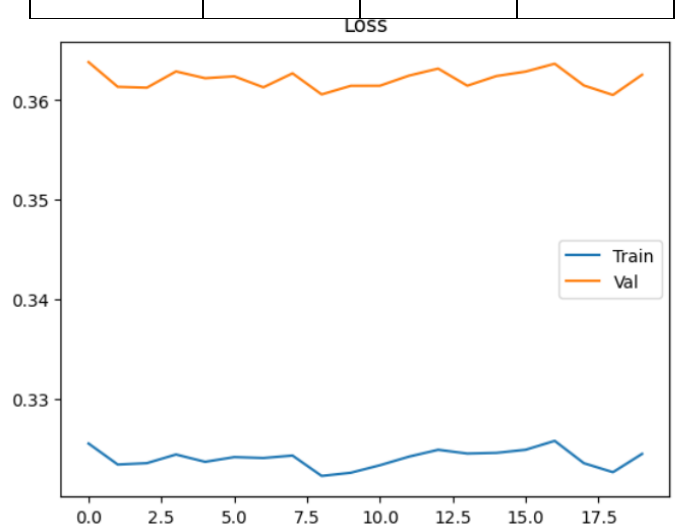


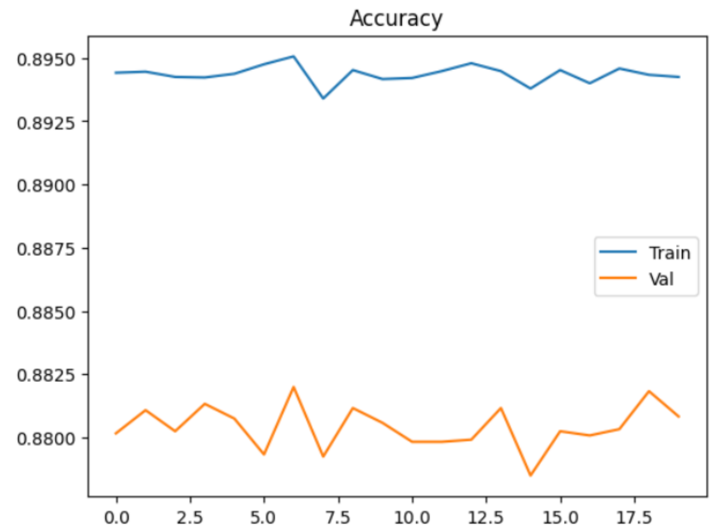
Random batch train:

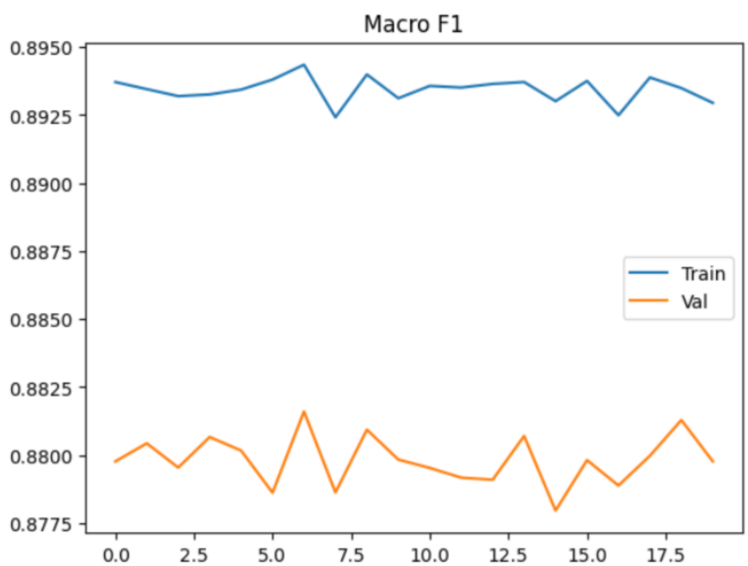
```
Model 1. 2
alpha:  0.001 decay: 0.999 dropout: 0.5 optimizer: ADAM  epoch: 20
```

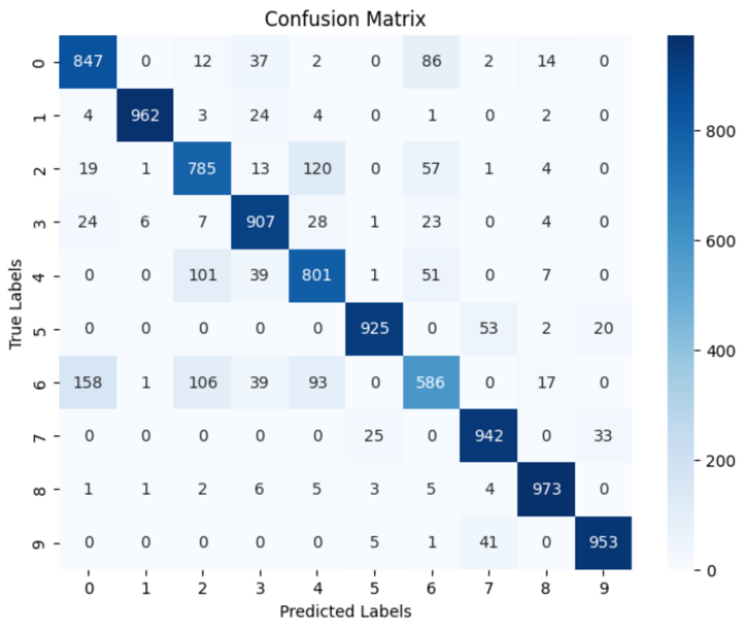
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.89425	0.324515	0.892952
Validation	0.880833	0.362561	0.879772
Test	0.8681	0.391785	0.866448




Accuracy
0.8950
0.8925
0.8900
0.8875
0.8850
0.8825
0.8800
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Train
Val


Macro F1
0.8950
0.8925
0.8900
0.8875
0.8850
0.8825
0.8800
0.8775
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Train
Val

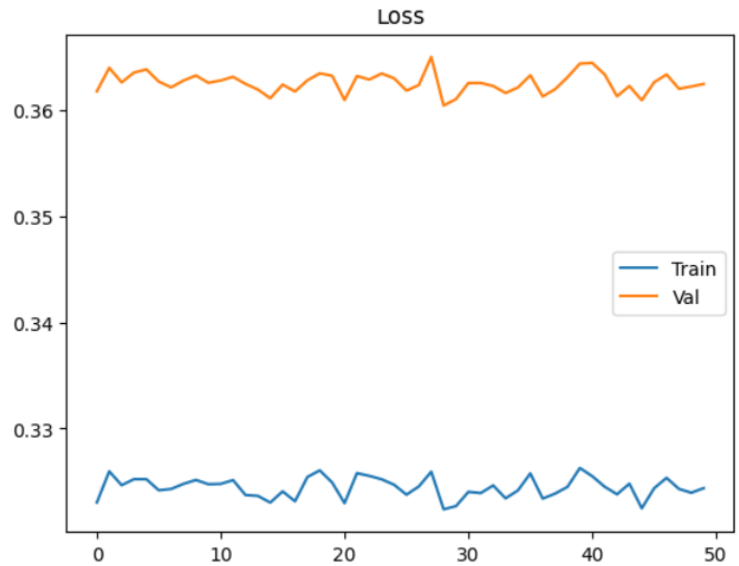


Random2 batch train:

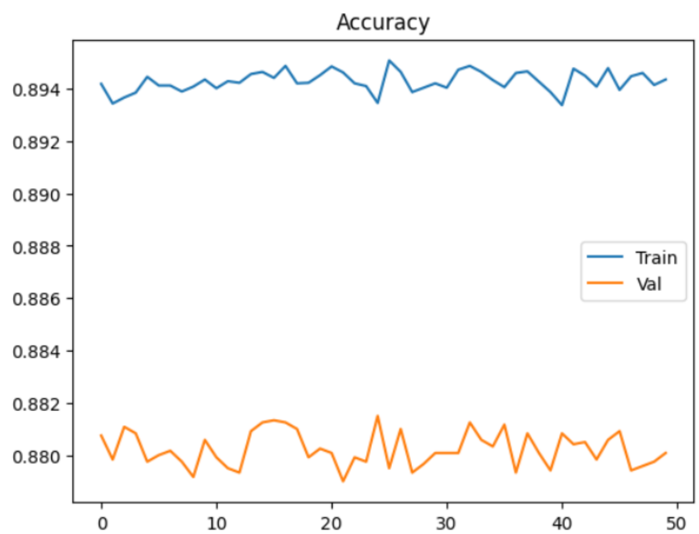
```
Model 1. 2
alpha:  0.001 decay: 0.999 dropout: 0.5 optimizer: ADAM  epoch: 50
```

Performance

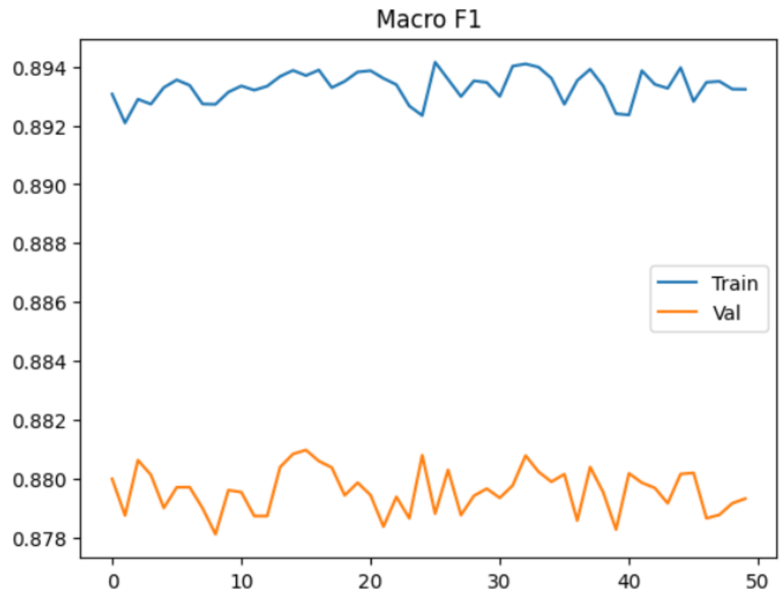
	Accuracy	Loss	Macro-F1-score
Train	0.894354	0.324370	0.8932335
Validation	0.880083	0.362511	0.879308
Test	0.8685	0.390367	0.867144

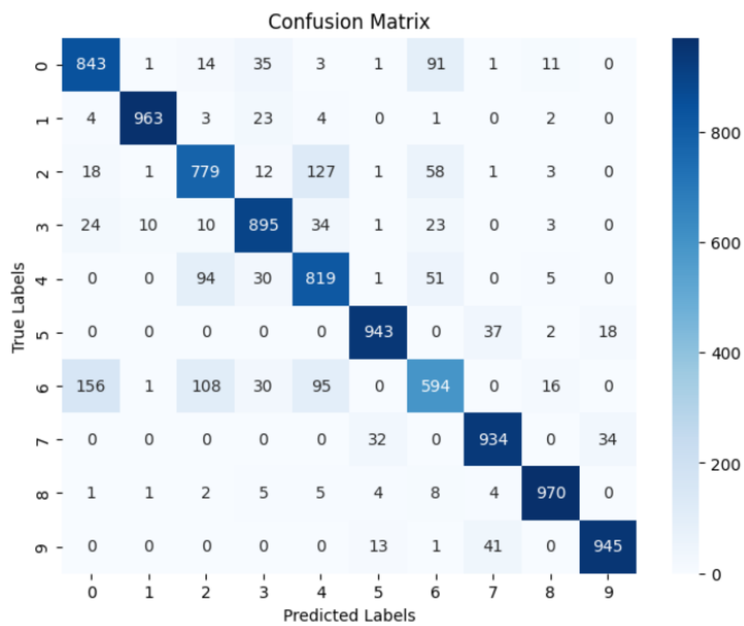


Loss
0.36
0.35
0.34
0.33
0
10
20
30
40
50
Train
Val



Accuracy
0.894
0.892
0.890
0.888
0.886
0.884
0.882
0.880
0
10
20
30
40
50
Train
Val





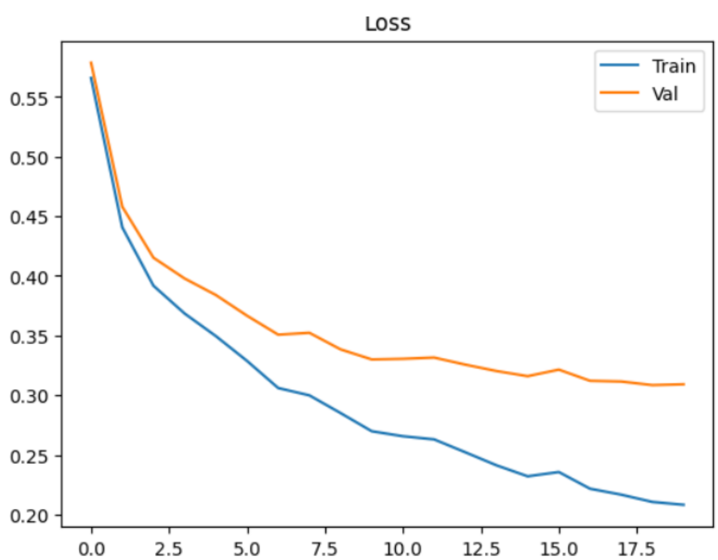
	0	1	2	3	4	5	6	7	8	9
0	843	1	14	35	3	1	91	1	11	0
1	4	963	3	23	4	0	1	0	2	0
2	18	1	779	12	127	1	58	1	3	0
3	24	10	10	895	34	1	23	0	3	0
4	0	0	94	30	819	1	51	0	5	0
5	0	0	0	0	0	943	0	37	2	18
6	156	1	108	30	95	0	594	0	16	0
7	0	0	0	0	0	32	0	934	0	34
8	1	1	2	5	5	4	8	4	970	0
9	0	0	0	0	0	13	1	41	0	945

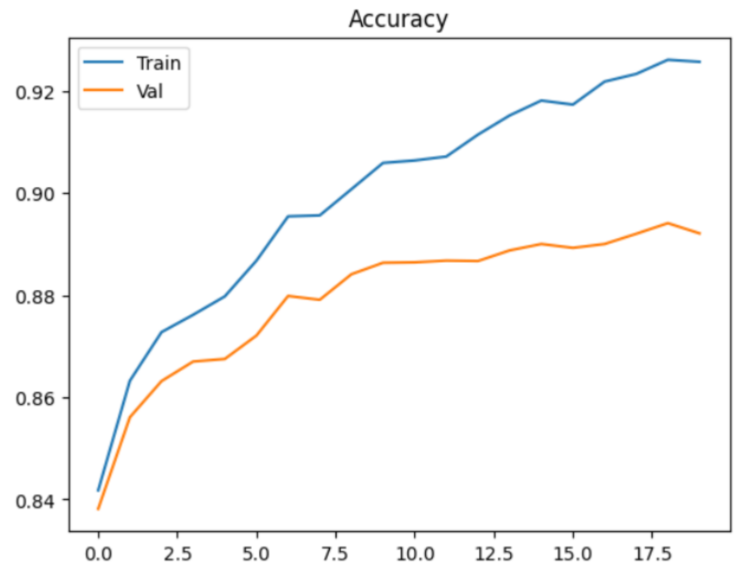
Sequential batch train:

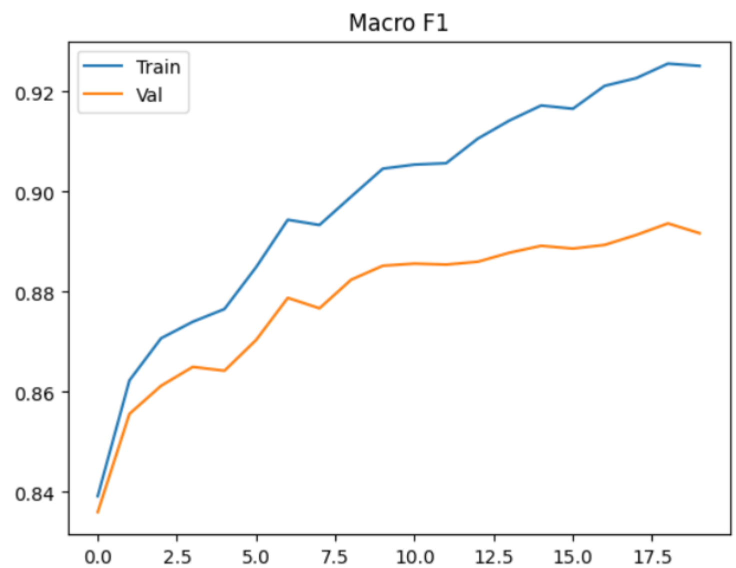
```
Model 1. 3
alpha:  0.00078125 decay: 0.9999 dropout: 0.5 optimizer: ADAM  epoch: 20
```

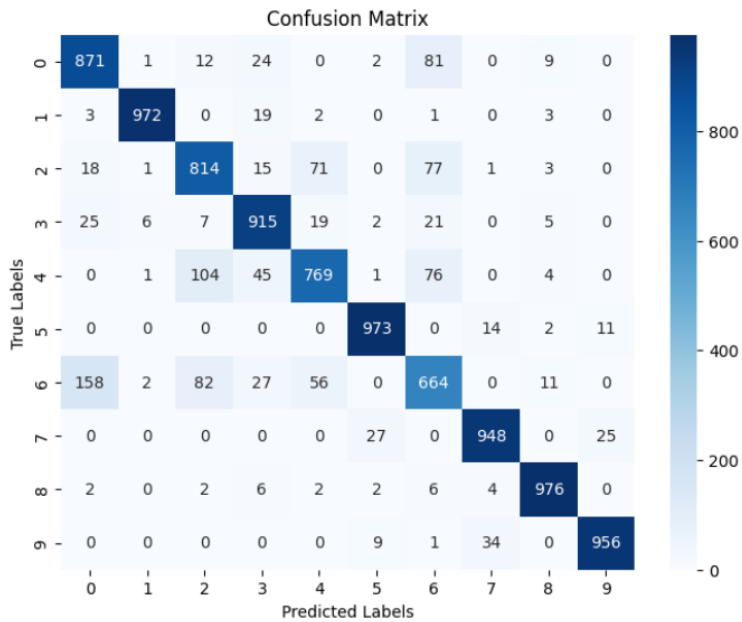
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.925666	0.208230	0.92515
Validation	0.892083	0.3092028	0.891701
Test	0.8858	0.336379	0.884978




Accuracy
Train
Val
0.92
0.90
0.88
0.86
0.84
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Macro F1
Train
Val
0.92
0.90
0.88
0.86
0.84
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

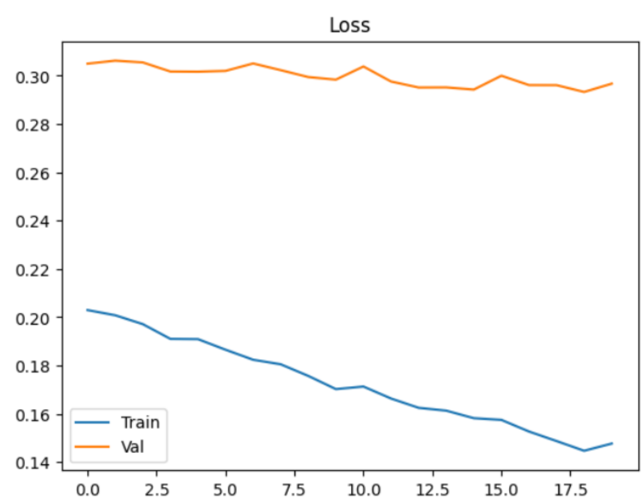


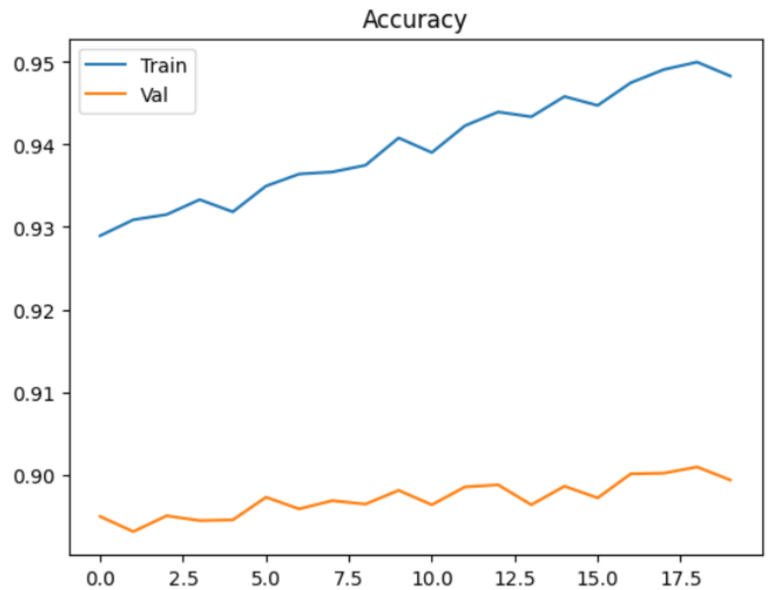
Random batch train:

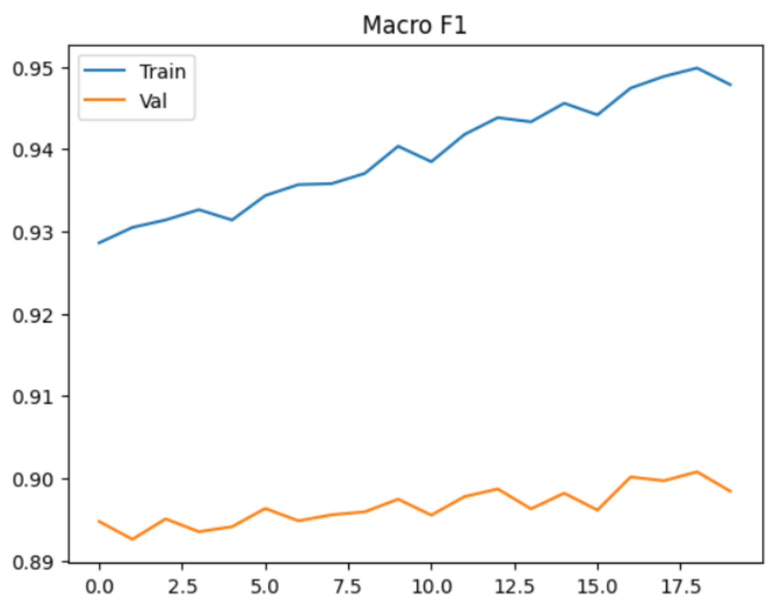
```
Model 1. 3
alpha:  0.00078125 decay: 0.9999 dropout: 0.5 optimizer: ADAM epoch: 20
```

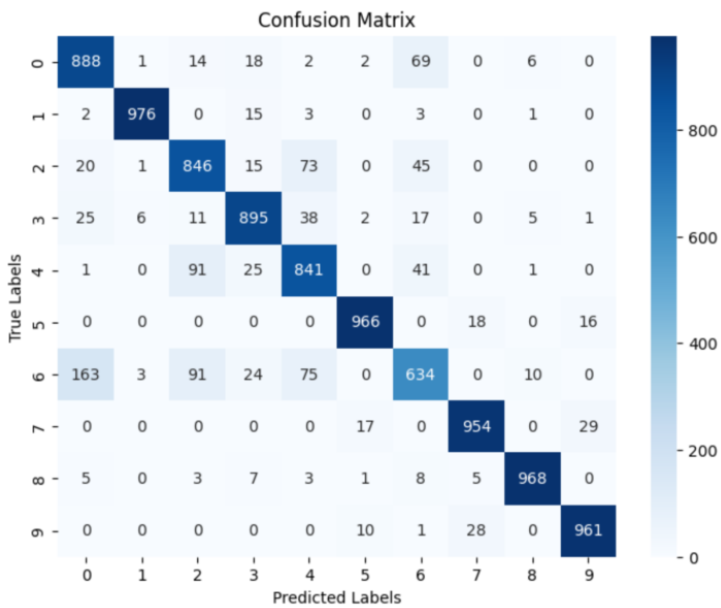
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.948292	0.147619	0.947818
Validation	0.899333	0.296713	0.898418
Test	0.8929	0.319253	0.891751




Accuracy
Train
Val
0.95
0.94
0.93
0.92
0.91
0.90
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Macro F1
Train
Val
0.95
0.94
0.93
0.92
0.91
0.90
0.89
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

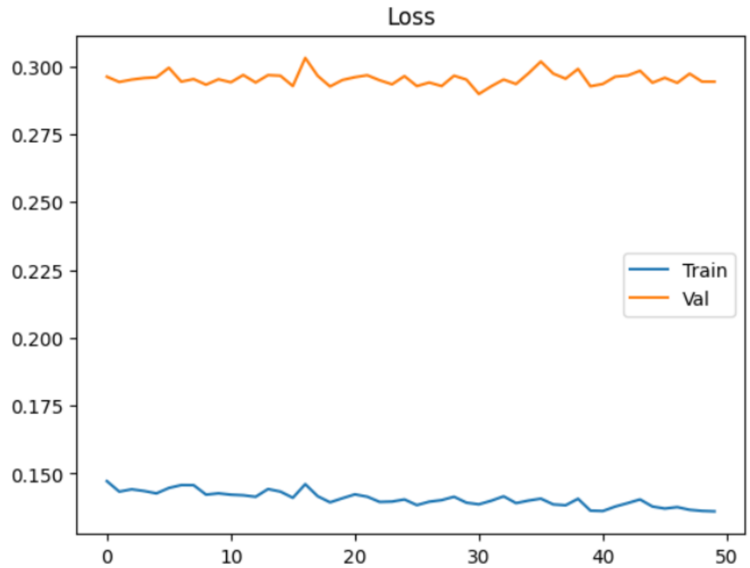


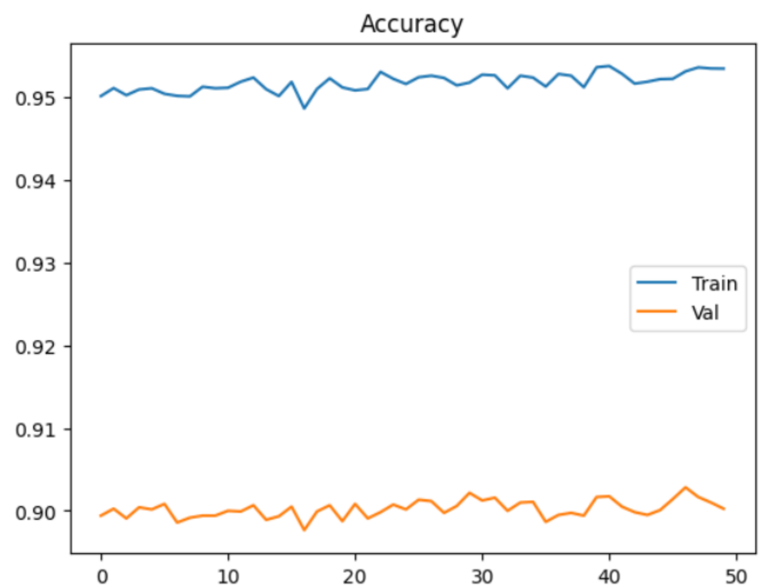
Random2 batch train:

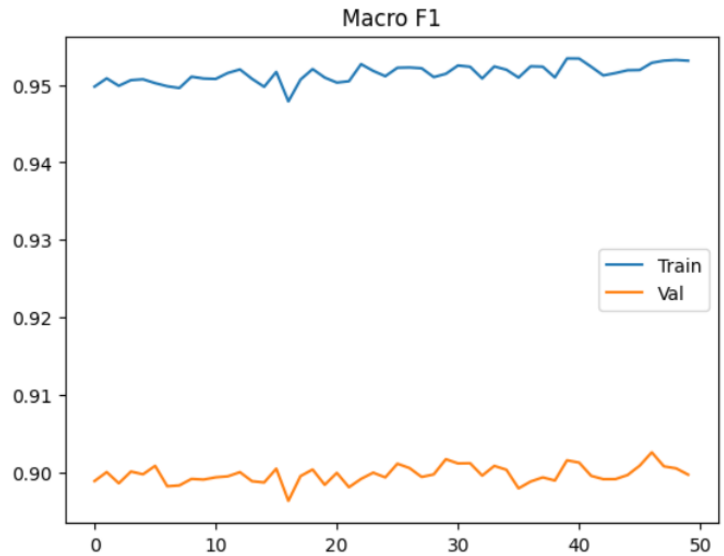
```
Model 1.3
alpha:  0.00078125 decay: 0.9999 dropout: 0.5 optimizer: ADAM  epoch: 50
```

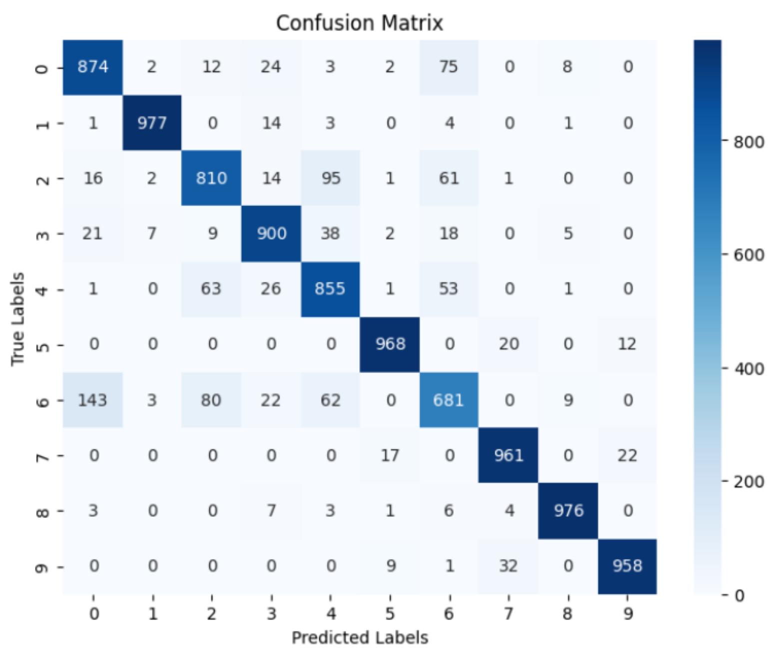
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.953375	0.136085	0.953122
Validation	0.90025	0.294294	0.89973
Test	0.896	0.312989	0.895341


Loss
0.300
0.275
0.250
0.225
0.200
0.175
0.150
0
10
20
30
40
50
Train
Val


Accuracy
0.95
0.94
0.93
0.92
0.91
0.90
0
10
20
30
40
50
Train
Val


Macro F1
0.95
0.94
0.93
0.92
0.91
0.90
0
10
20
30
40
50
Train
Val


Confusion Matrix
874 2 12 24 3 2 75 0 8 0
1 977 0 14 3 0 4 0 1 0
16 2 810 14 95 1 61 1 0 0
21 7 9 900 38 2 18 0 5 0
1 0 63 26 855 1 53 0 1 0
0 0 0 0 0 968 0 20 0 12
143 3 80 22 62 0 681 0 9 0
0 0 0 0 0 17 0 961 0 22
3 0 0 7 3 1 6 4 976 0
0 0 0 0 0 9 1 32 0 958
True Labels
Predicted Labels
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

Model 2

#layer 1

Dense(784, 392)

BatchNormalization()

Relu()

Dropout()

#layer 2

Dense(392, 196)

BatchNormalization()

Relu()

Dropout()

#layer 3

Dense(196, 10)

BatchNormalization()

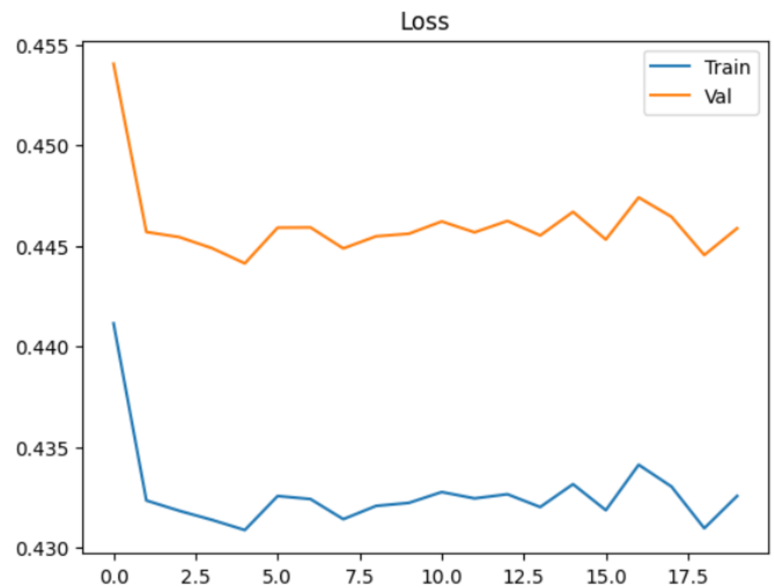
Softmax()

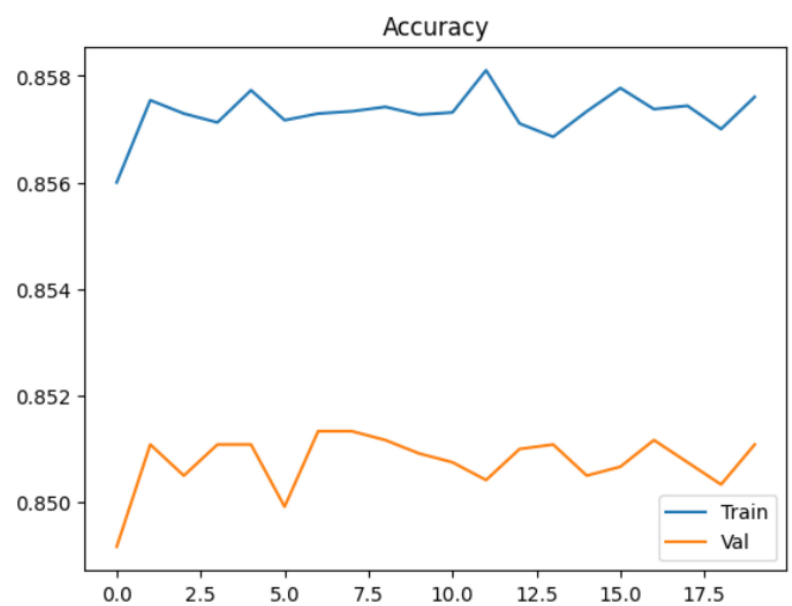
Sequential batch train:

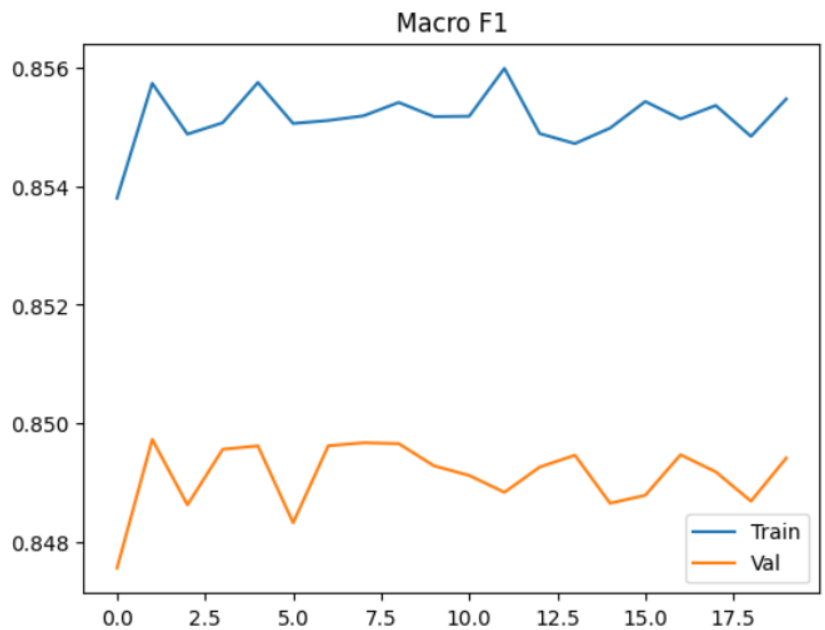
```
Model 2. 1
alpha:  0.01 decay: 0.99 dropout: 0.5 optimizer: ADAM  epoch: 20
```

Performance

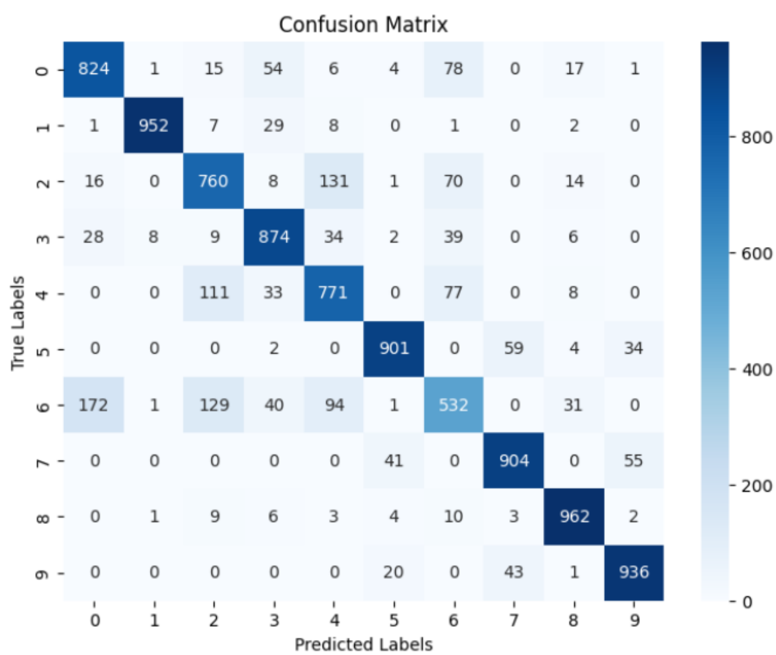
	Accuracy	Loss	Macro-F1-score
Train	0.857604	0.43256	0.855466
Validation	0.851083	0.445872	0.849415
Test	0.8416	0.46996	0.839568


Loss
Train
Val
0.455
0.450
0.445
0.440
0.435
0.430
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Accuracy
Train
Val
0.858
0.856
0.854
0.852
0.850
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5



Macro F1
0.856
0.854
0.852
0.850
0.848
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Train
Val



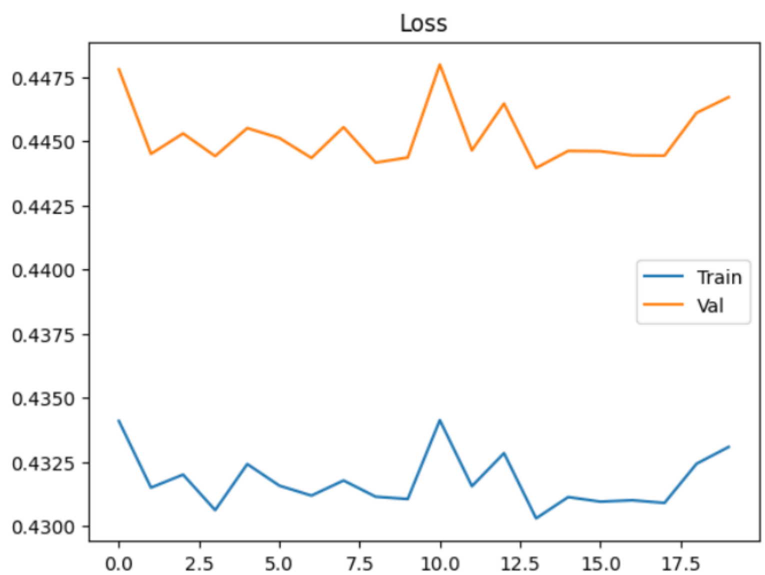
Confusion Matrix
True Labels
Predicted Labels
824 1 15 54 6 4 78 0 17 1
1 952 7 29 8 0 1 0 2 0
16 0 760 8 131 1 70 0 14 0
28 8 9 874 34 2 39 0 6 0
0 0 111 33 771 0 77 0 8 0
0 0 0 2 0 901 0 59 4 34
172 1 129 40 94 1 532 0 31 0
0 0 0 0 0 41 0 904 0 55
0 1 9 6 3 4 10 3 962 2
0 0 0 0 0 20 0 43 1 936
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

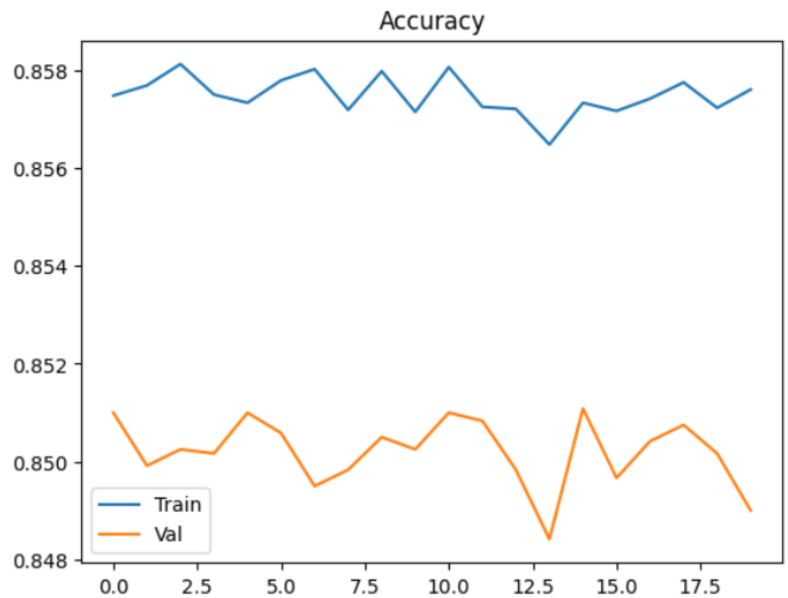
Random batch train:

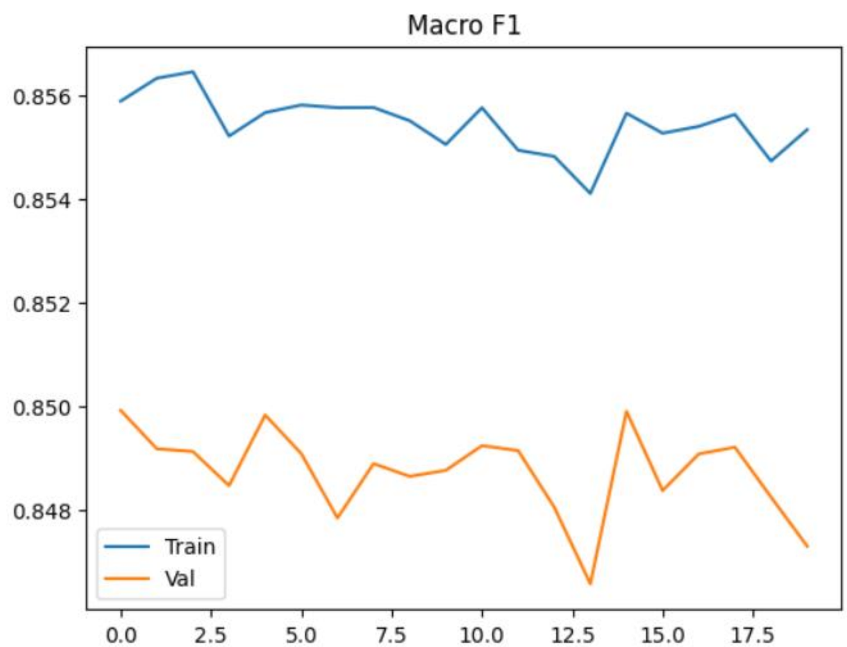
```
Model 2. 1
alpha:  0.01 decay: 0.99 dropout: 0.5 optimizer: ADAM  epoch: 20
```

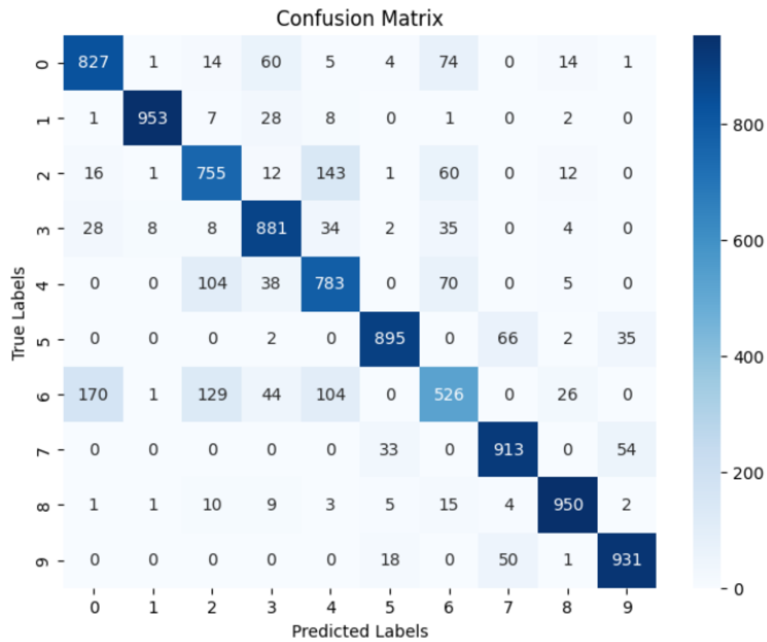
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.857604	0.4330816	0.855339
Validation	0.849	0.446709	0.847290
Test	0.8414	0.47042	0.839289




Accuracy
0.858
0.856
0.854
0.852
0.850
0.848
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Train
Val


Macro F1
0.856
0.854
0.852
0.850
0.848
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Train
Val

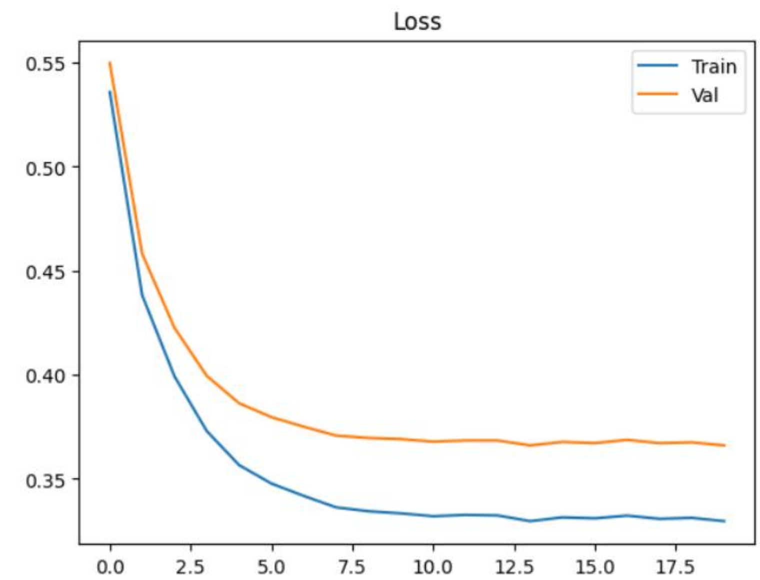


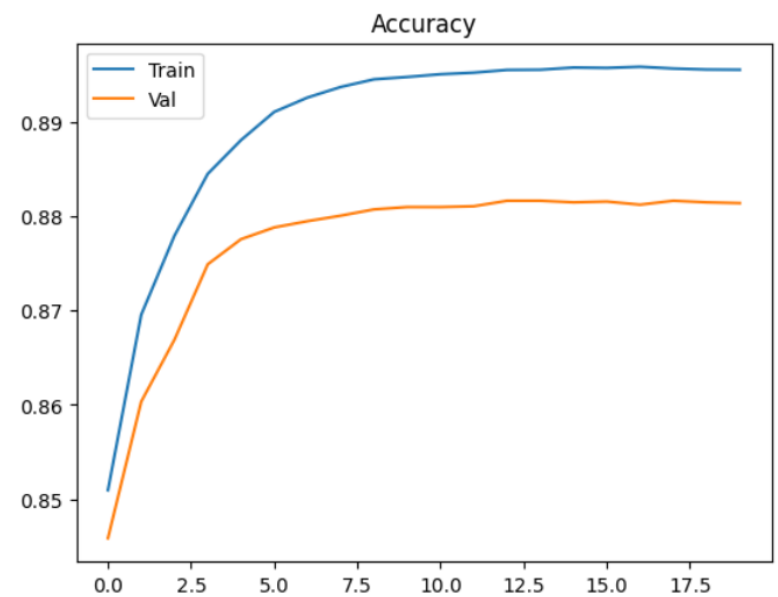
Sequential batch train:

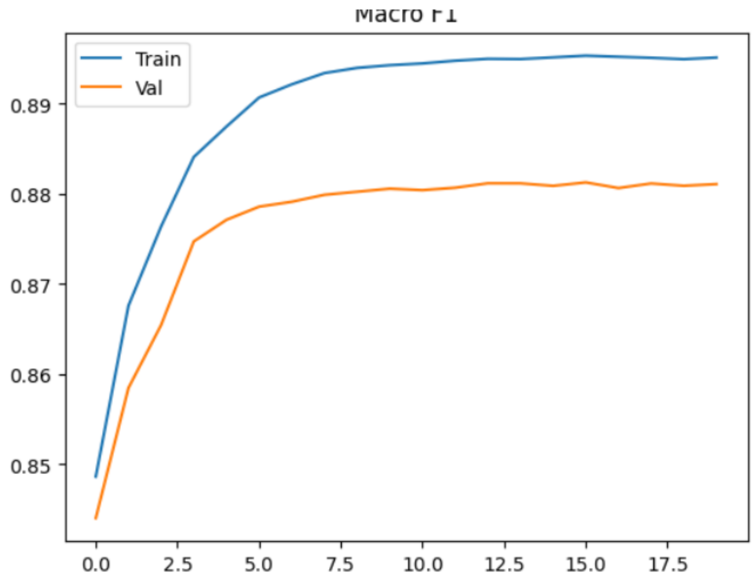
```
Model 2. 2
alpha:  0.001 decay: 0.999 dropout: 0.5 optimizer: YOGI  epoch: 20
```

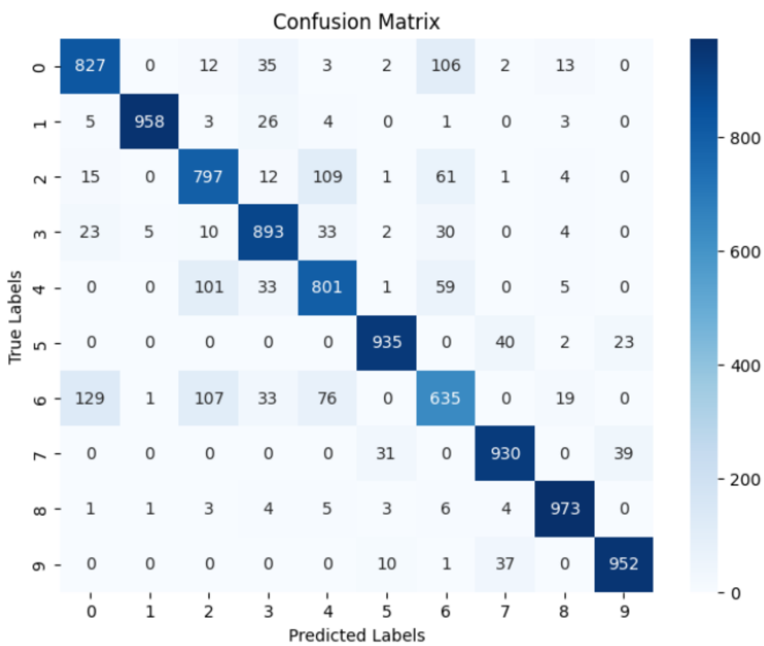
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.895583	0.329672	0.895132
Validation	0.881416	0.366073	0.881093
Test	0.8701	0.393298	0.86938


Loss
Train
Val
0.55
0.50
0.45
0.40
0.35
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Accuracy
Train
Val
0.89
0.88
0.87
0.86
0.85
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5



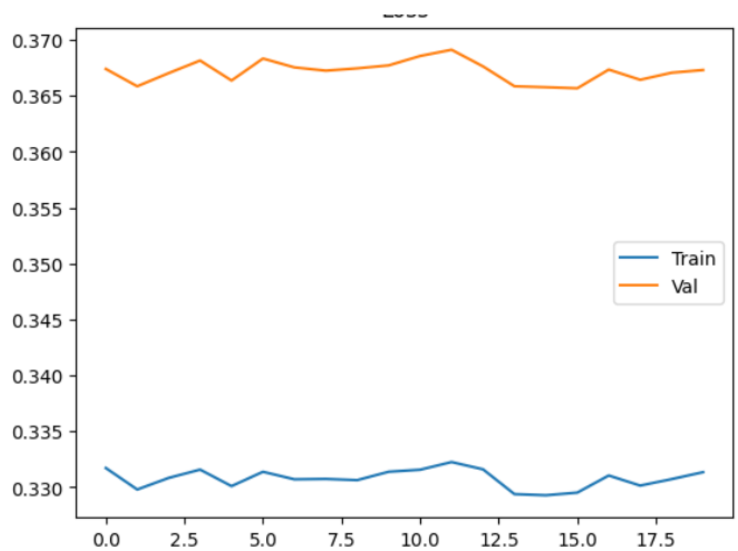


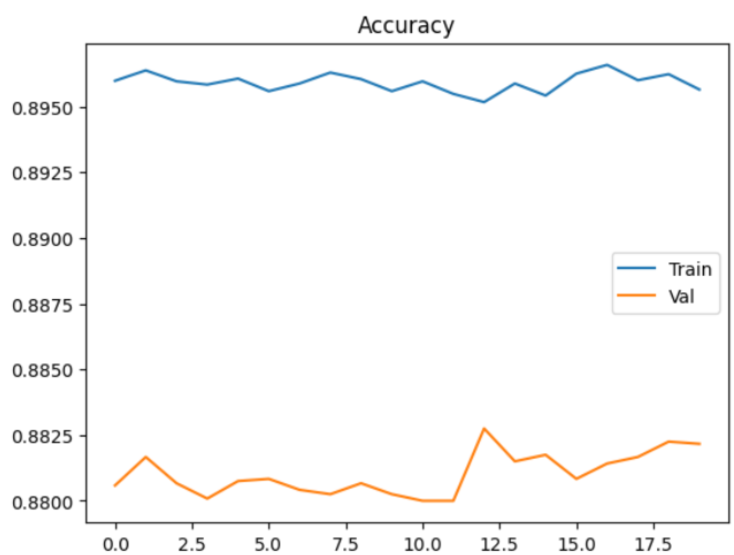
Random batch train:

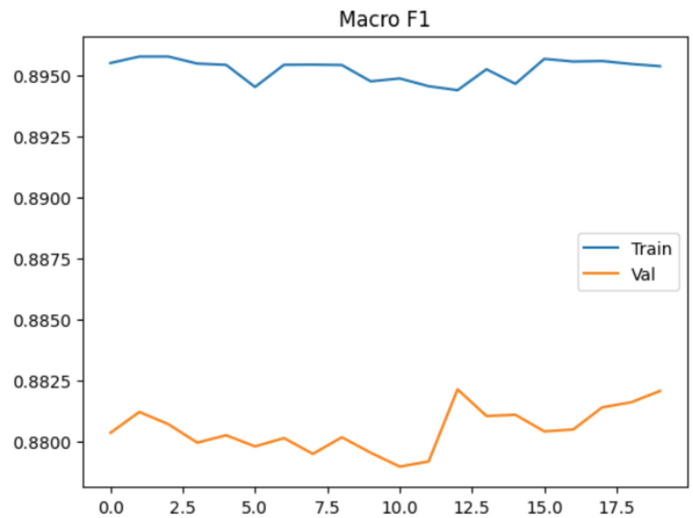
```
Model 2. 2
alpha:  0.001 decay: 0.999 dropout: 0.5 optimizer: YOGI  epoch: 20
```

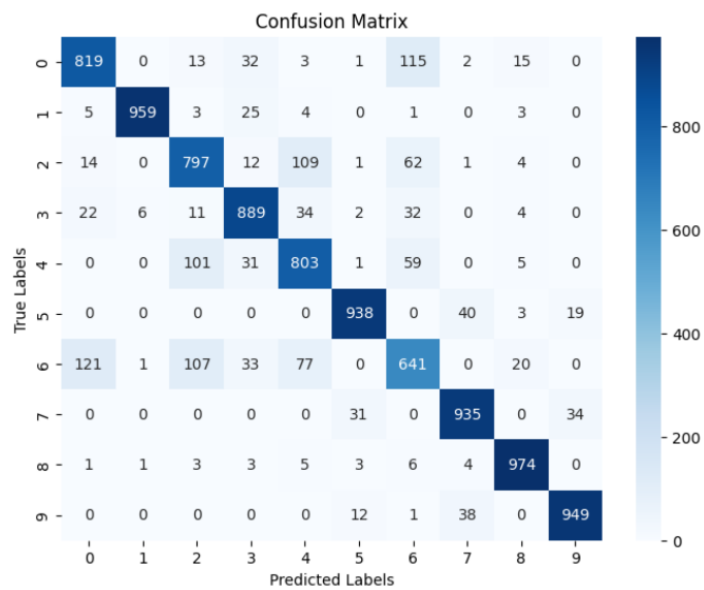
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.89564	0.33135	0.895393
Validation	0.882166	0.367309	0.882061
Test	0.8704	0.394747	0.86983







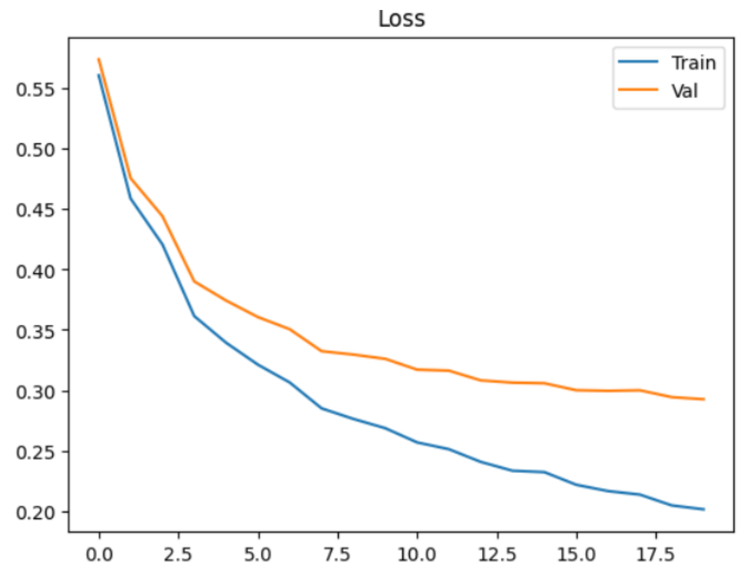


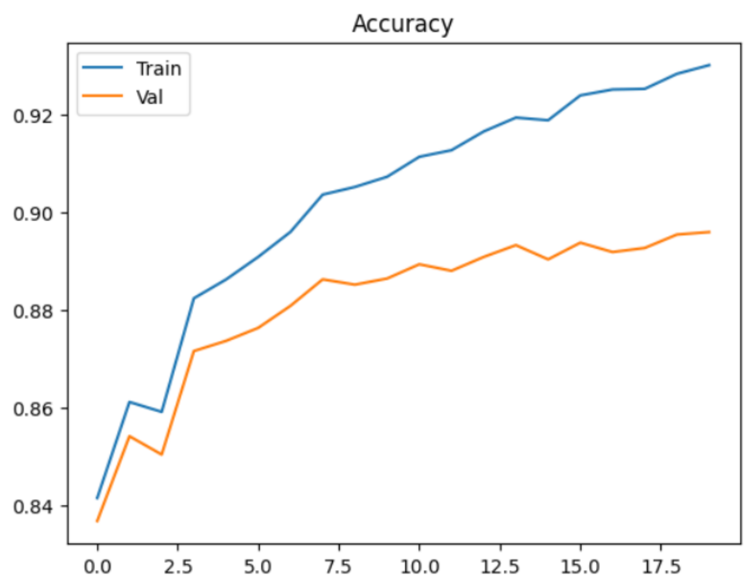
Sequential batch train:

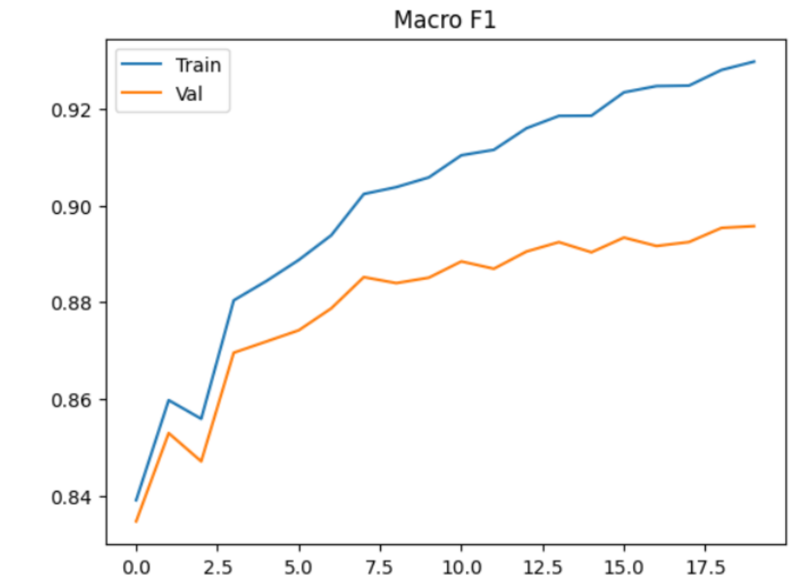
```
Model 2. 3
alpha:  0.00078125 decay: 0.9999 dropout: 0.5 optimizer: YOGI  epoch: 20
```

Performance

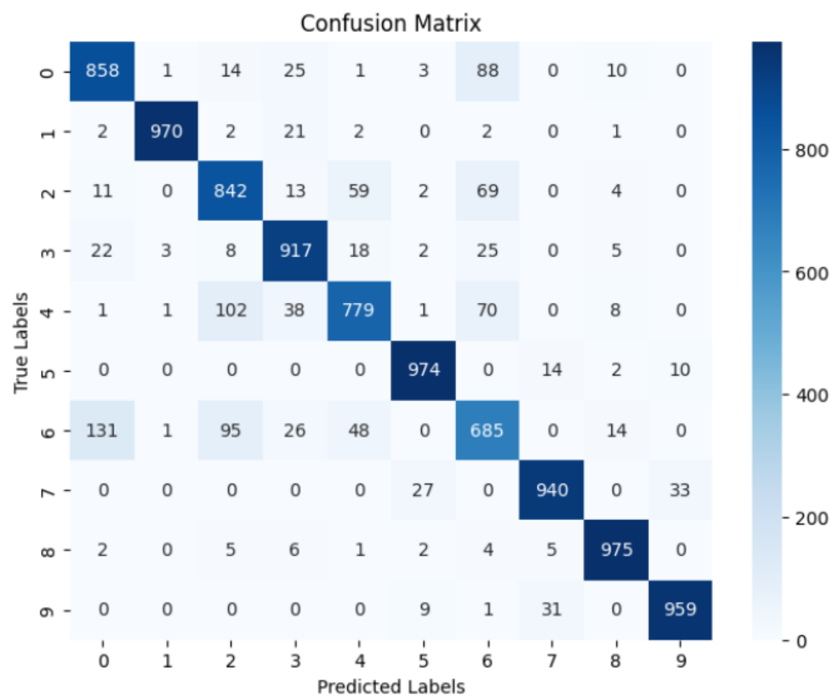
	Accuracy	Loss	Macro-F1-score
Train	0.93006	0.201376	0.929762
Validation	0.895916	0.292543	0.895729
Test	0.8899	0.31887	0.88932


Loss
Train
Val
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Accuracy
Train
Val
0.92
0.90
0.88
0.86
0.84
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5



Macro F1
Train
Val
0.92
0.90
0.88
0.86
0.84
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5



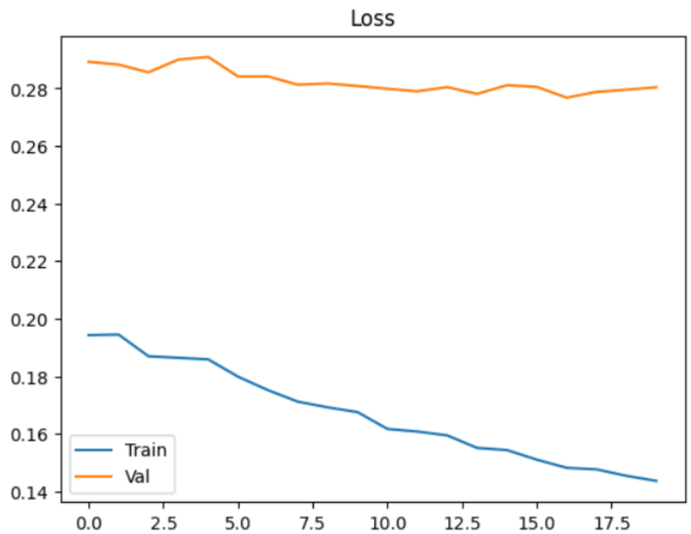
Confusion Matrix
True Labels
Predicted Labels
858 1 14 25 1 3 88 0 10 0
2 970 2 21 2 0 2 0 1 0
11 0 842 13 59 2 69 0 4 0
22 3 8 917 18 2 25 0 5 0
1 1 102 38 779 1 70 0 8 0
0 0 0 0 0 974 0 14 2 10
131 1 95 26 48 0 685 0 14 0
0 0 0 0 0 27 0 940 0 33
2 0 5 6 1 2 4 5 975 0
0 0 0 0 0 9 1 31 0 959
800
600
400
200
0

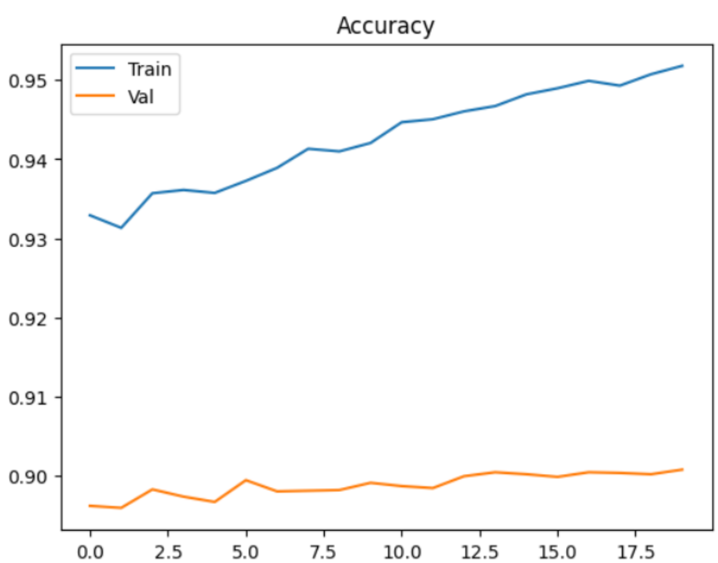
Random batch train:

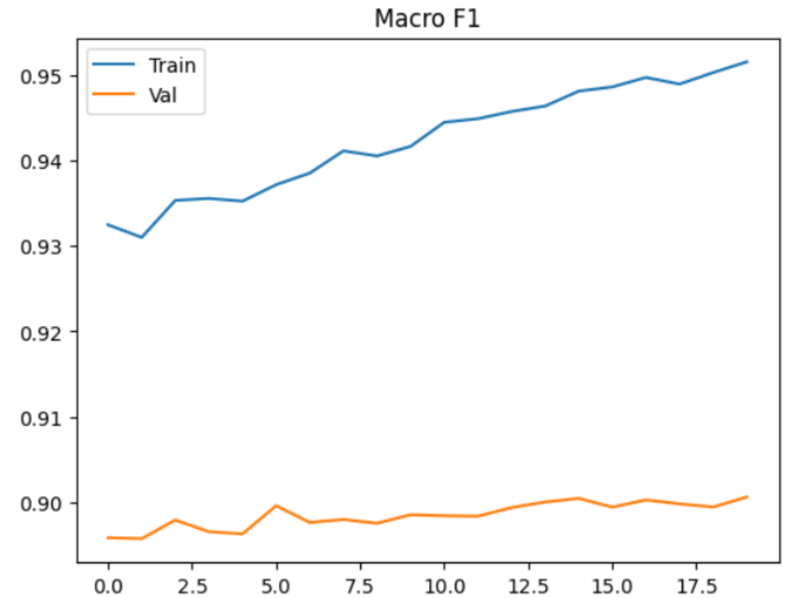
```
Model 2. 3
alpha:  0.00078125 decay: 0.9999 dropout: 0.5 optimizer: YOGI  epoch: 20
```

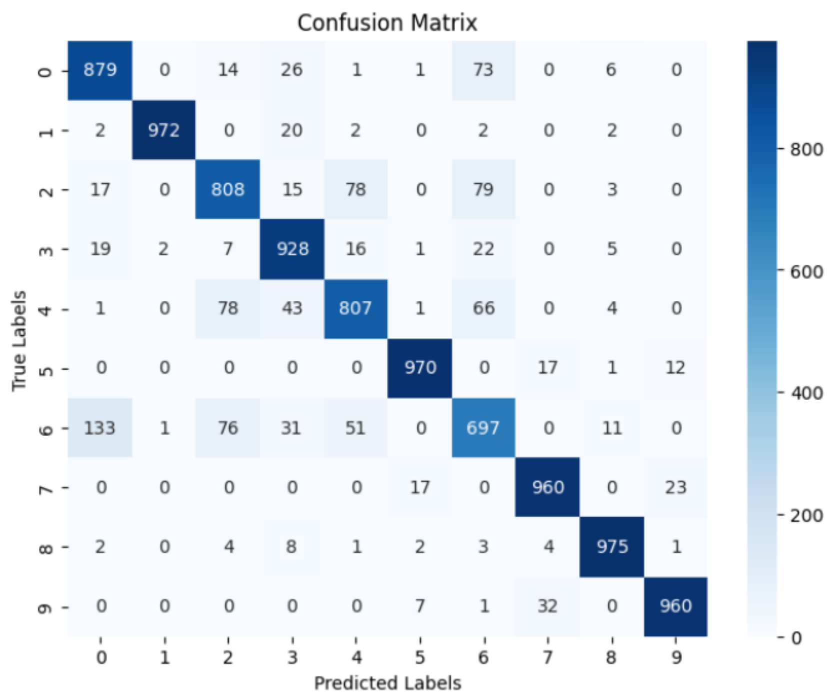
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.951791	0.143686	0.951582
Validation	0.90075	0.280352	0.90059
Test	0.8956	0.300483	0.895049






Macro F1
Train
Val
0.95
0.94
0.93
0.92
0.91
0.90
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Confusion Matrix
True Labels
Predicted Labels
0 879 0 14 26 1 1 73 0 6 0
1 2 972 0 20 2 0 2 0 2 0
2 17 0 808 15 78 0 79 0 3 0
3 19 2 7 928 16 1 22 0 5 0
4 1 0 78 43 807 1 66 0 4 0
5 0 0 0 0 0 970 0 17 1 12
6 133 1 76 31 51 0 697 0 11 0
7 0 0 0 0 0 17 0 960 0 23
8 2 0 4 8 1 2 3 4 975 1
9 0 0 0 0 0 7 1 32 0 960
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

\

Model 3

#layer 1

Dense(784, 392)

Relu()

BatchNormalization()

Dropout()

#layer 2

Dense(392, 196)

Relu()

BatchNormalization()

Dropout()

#layer 3

Dense(196, 10)

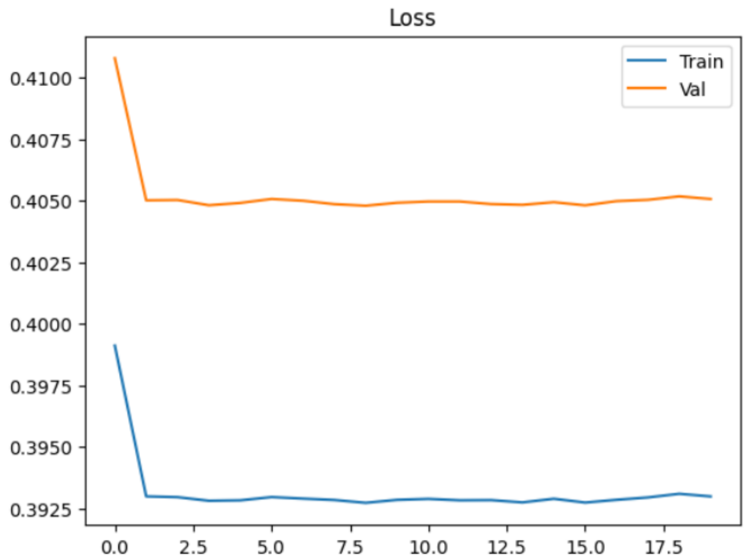
Softmax()

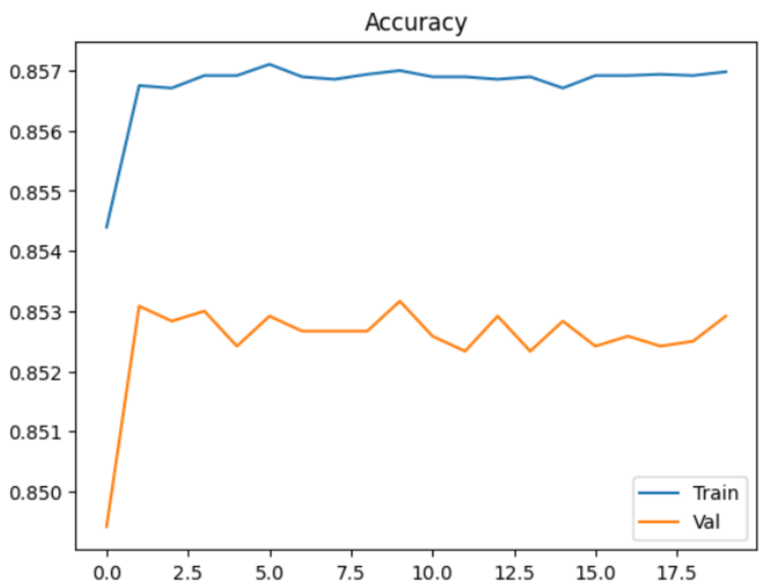
Sequential batch train:

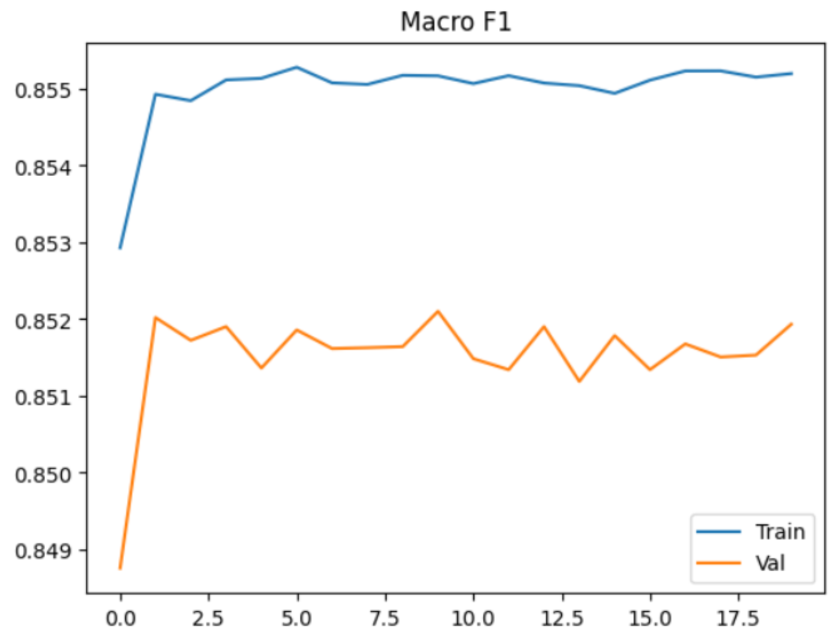
```
Model 3. 1
alpha:  0.01 decay: 0.99 dropout: 0.5 optimizer: ADAM  epoch: 20
```

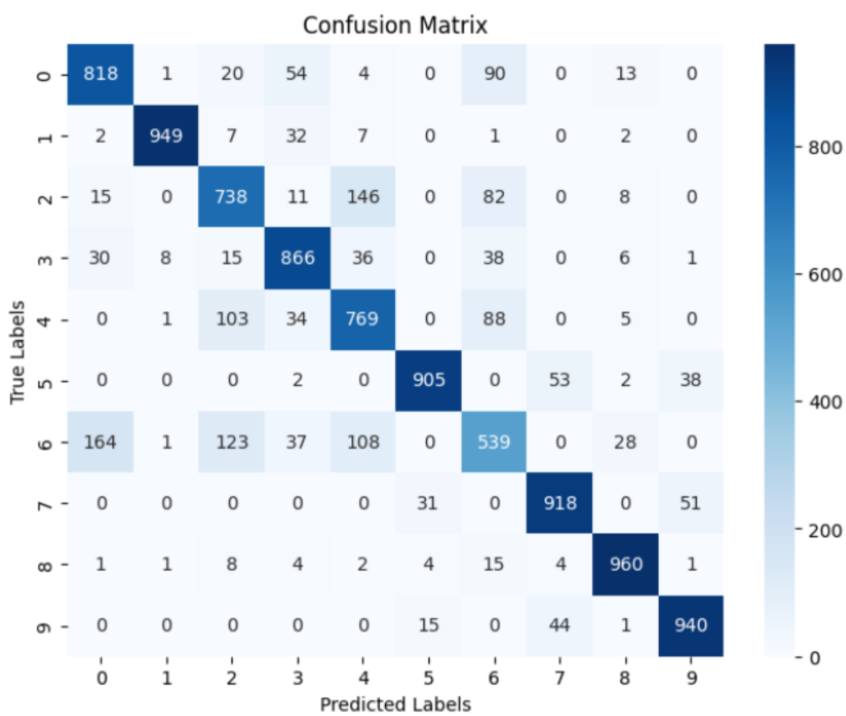
Performance

	Accuracy	**Loss**	**Macro-F1-score**
Train	0.856979	0.392987	0.855197
Validation	0.852916	0.405074	0.851933
Test	0.8402	0.436020	0.83887


Loss
Train
Val
0.4100
0.4075
0.4050
0.4025
0.4000
0.3975
0.3950
0.3925
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Accuracy
0.857
0.856
0.855
0.854
0.853
0.852
0.851
0.850
Train
Val
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Macro F1
0.855
0.854
0.853
0.852
0.851
0.850
0.849
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Train
Val

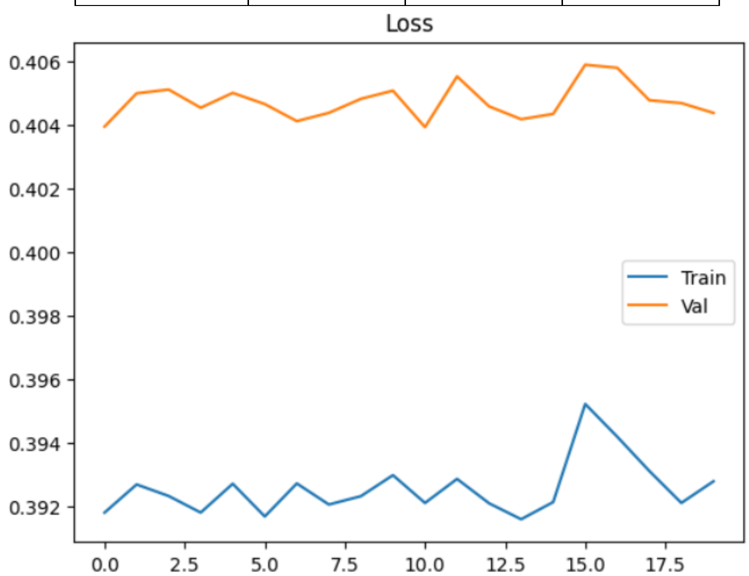

Confusion Matrix
True Labels
Predicted Labels
0 1 2 3 4 5 6 7 8 9
0: 818 1 20 54 4 0 90 0 13 0
1: 2 949 7 32 7 0 1 0 2 0
2: 15 0 738 11 146 0 82 0 8 0
3: 30 8 15 866 36 0 38 0 6 1
4: 0 1 103 34 769 0 88 0 5 0
5: 0 0 0 2 0 905 0 53 2 38
6: 164 1 123 37 108 0 539 0 28 0
7: 0 0 0 0 0 31 0 918 0 51
8: 1 1 8 4 2 4 15 4 960 1
9: 0 0 0 0 0 15 0 44 1 940
800
600
400
200
0

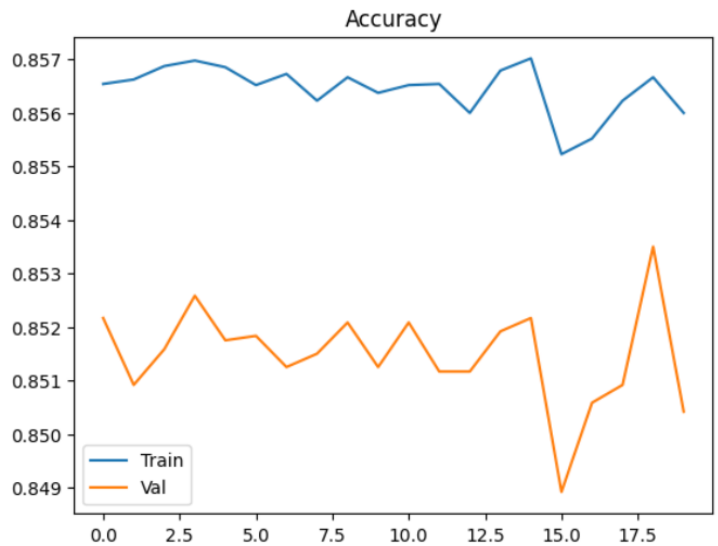
Random batch train:

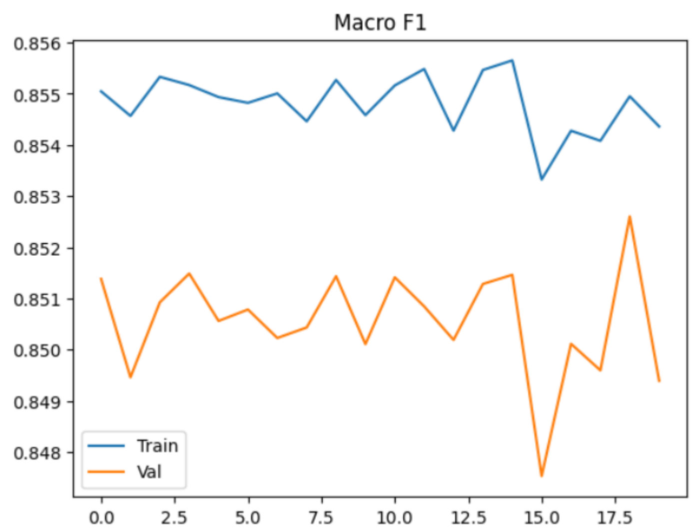
```
Model 3. 1
alpha:  0.01 decay: 0.99 dropout: 0.5 optimizer: ADAM  epoch: 20
```

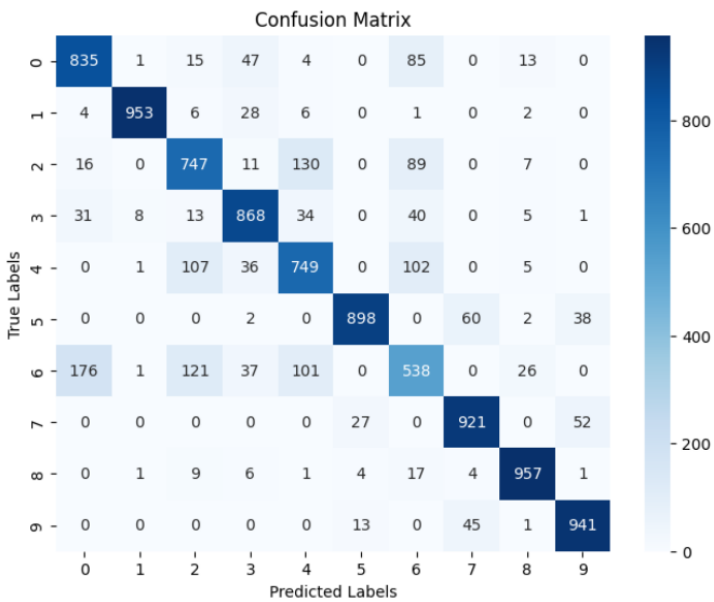
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.856	0.39277	0.85436
Validation	0.850416	0.40437	0.849391
Test	0.8407	0.433213	0.839421




Accuracy
0.857
0.856
0.855
0.854
0.853
0.852
0.851
0.850
0.849
Train
Val
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Macro F1
0.856
0.855
0.854
0.853
0.852
0.851
0.850
0.849
0.848
Train
Val
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

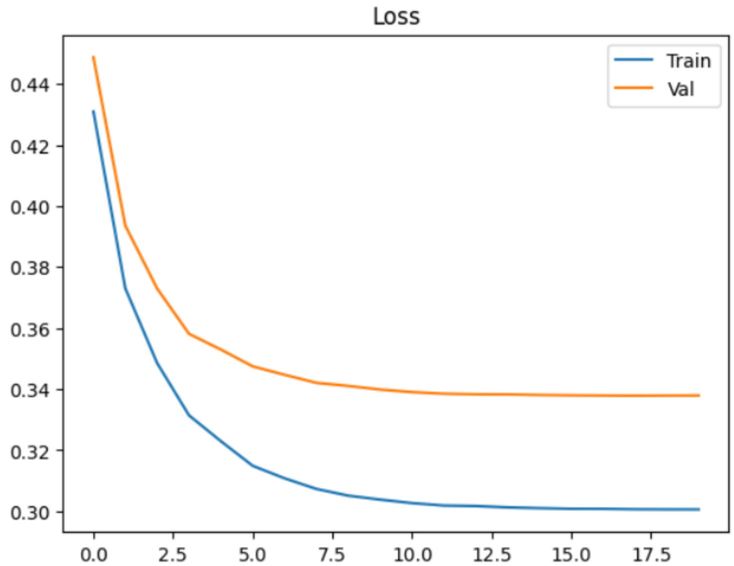


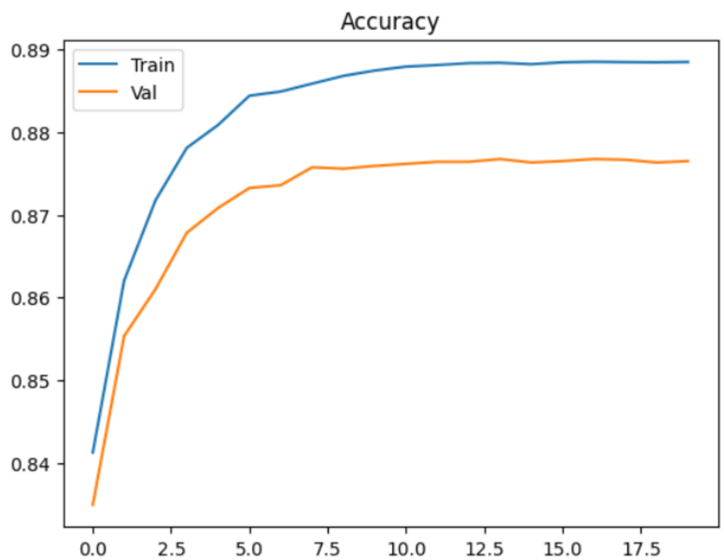
Sequential batch train:

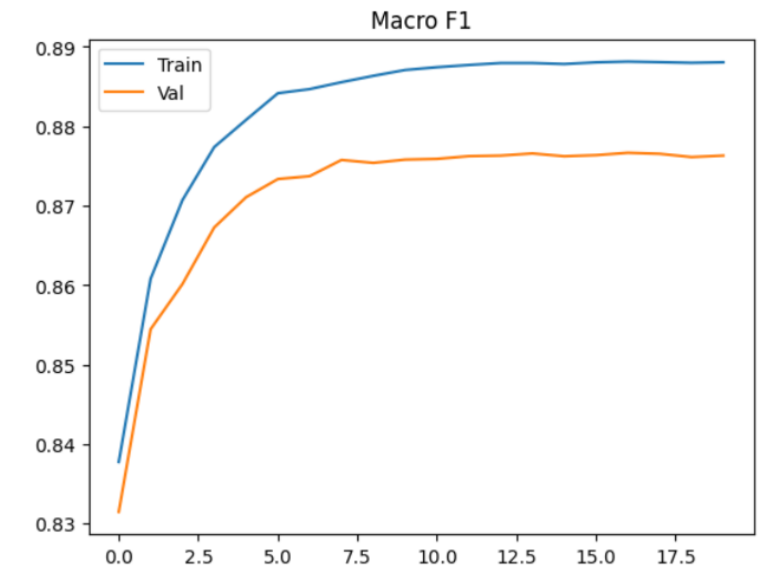
```
Model 3. 2
alpha:  0.001 decay: 0.999 dropout: 0.5 optimizer: ADAM  epoch: 20
```

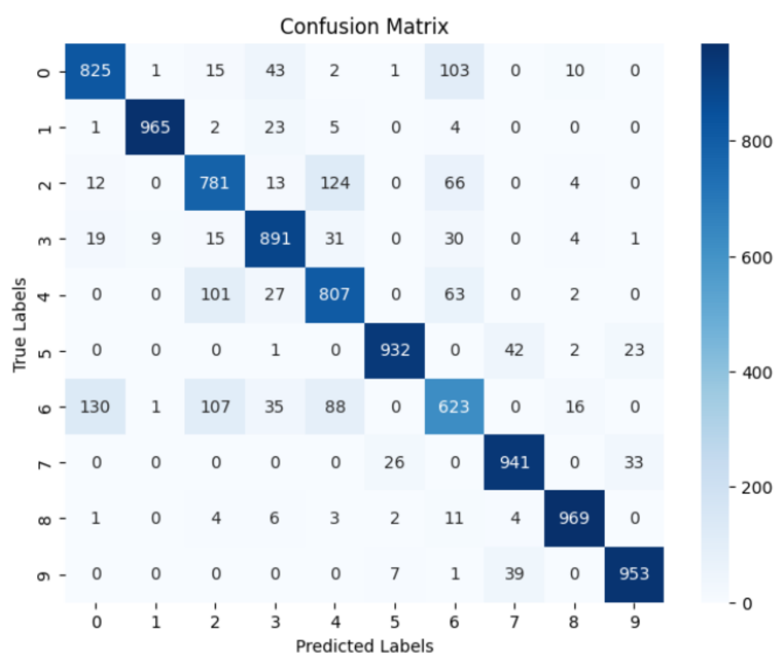
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.8885	0.300566	0.888054
Validation	0.8765	0.337894	0.876311
Test	0.8687	0.364276	0.868091


Loss
Train
Val
0.44
0.42
0.40
0.38
0.36
0.34
0.32
0.30
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Accuracy
Train
Val
0.89
0.88
0.87
0.86
0.85
0.84
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5





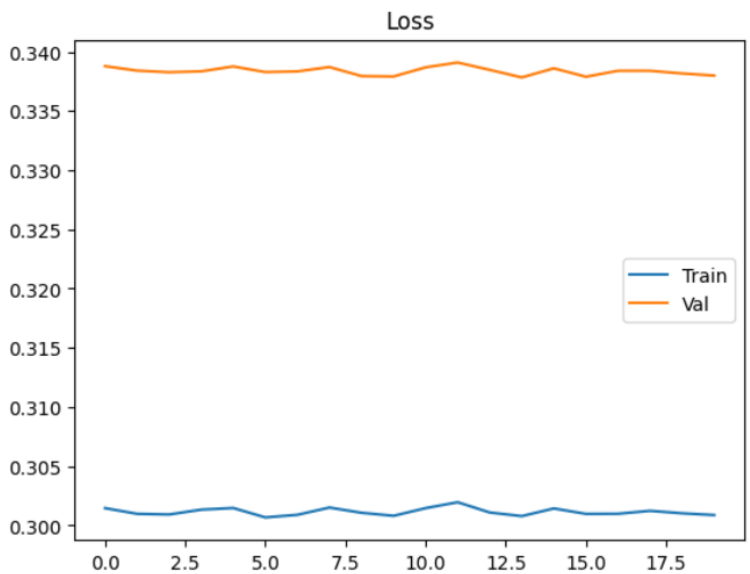
	0	1	2	3	4	5	6	7	8	9
0	825	1	15	43	2	1	103	0	10	0
1	1	965	2	23	5	0	4	0	0	0
2	12	0	781	13	124	0	66	0	4	0
3	19	9	15	891	31	0	30	0	4	1
4	0	0	101	27	807	0	63	0	2	0
5	0	0	0	1	0	932	0	42	2	23
6	130	1	107	35	88	0	623	0	16	0
7	0	0	0	0	0	26	0	941	0	33
8	1	0	4	6	3	2	11	4	969	0
9	0	0	0	0	0	7	1	39	0	953

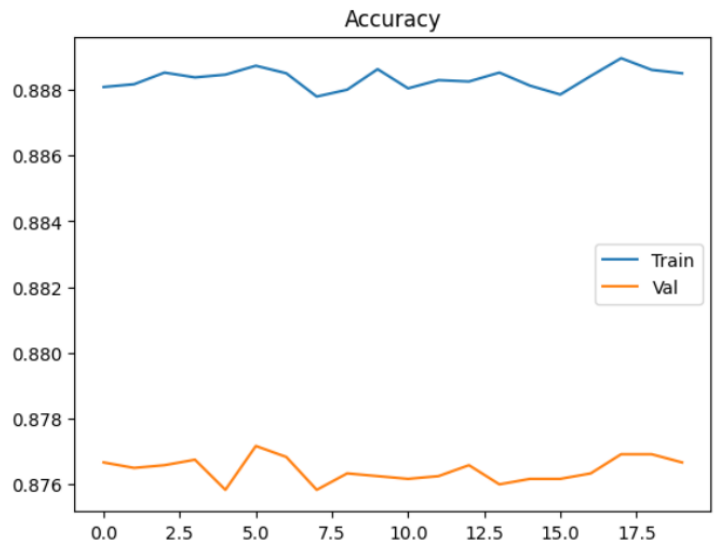
Random batch train:

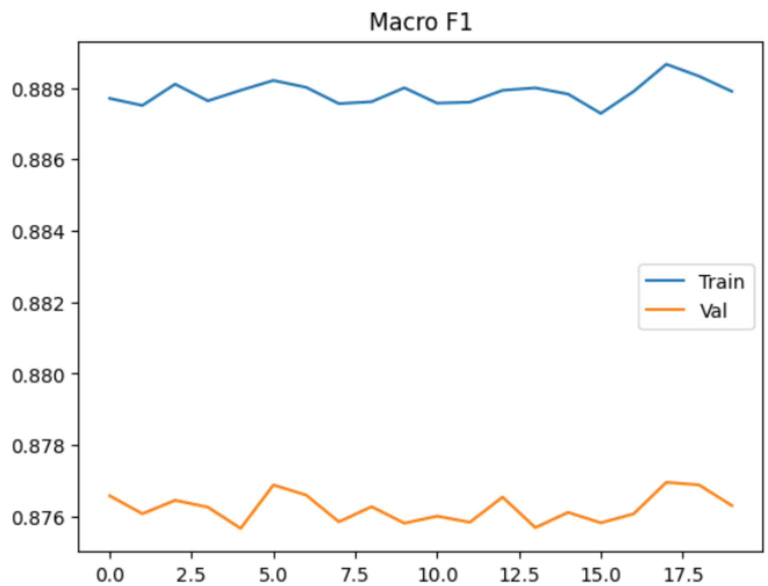
```
Model 3. 2
alpha:  0.001 decay: 0.999 dropout: 0.5 optimizer: ADAM  epoch: 20
```

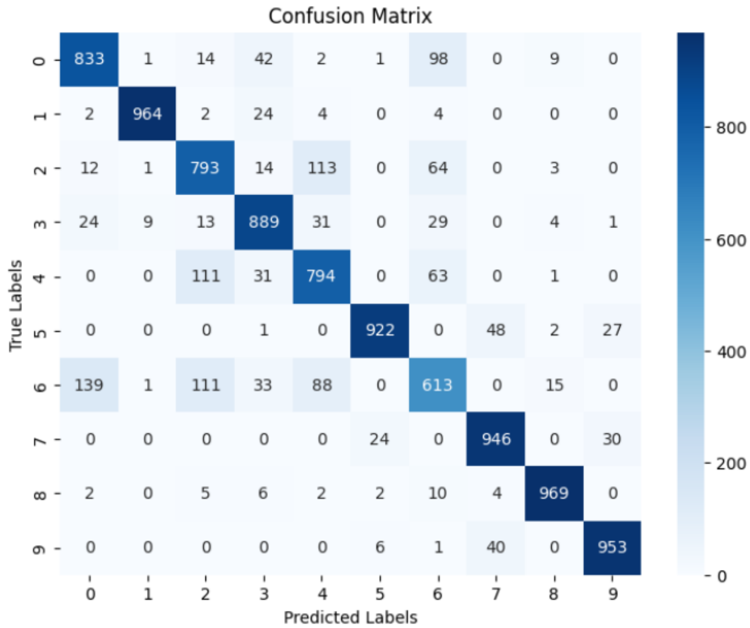
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.8885	0.300872	0.887904
Validation	0.876666	0.3379988	0.8763
Test	0.8676	0.364498	0.86684




Accuracy
0.888
0.886
0.884
0.882
0.880
0.878
0.876
Train
Val
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Macro F1
0.888
0.886
0.884
0.882
0.880
0.878
0.876
Train
Val
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

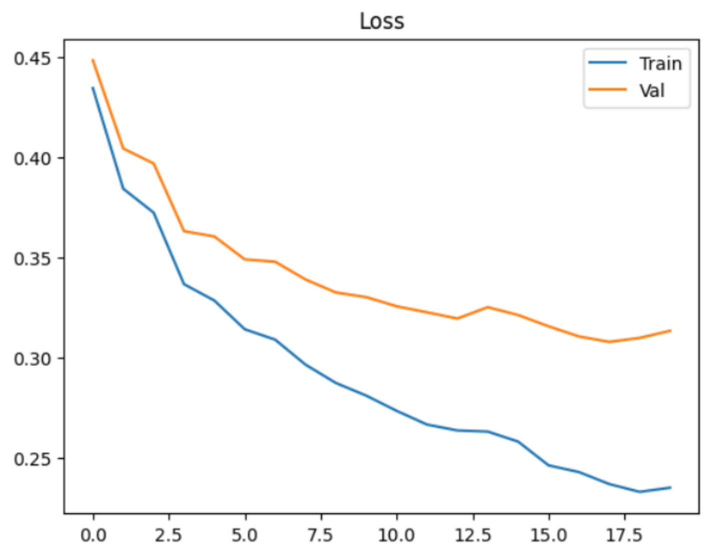


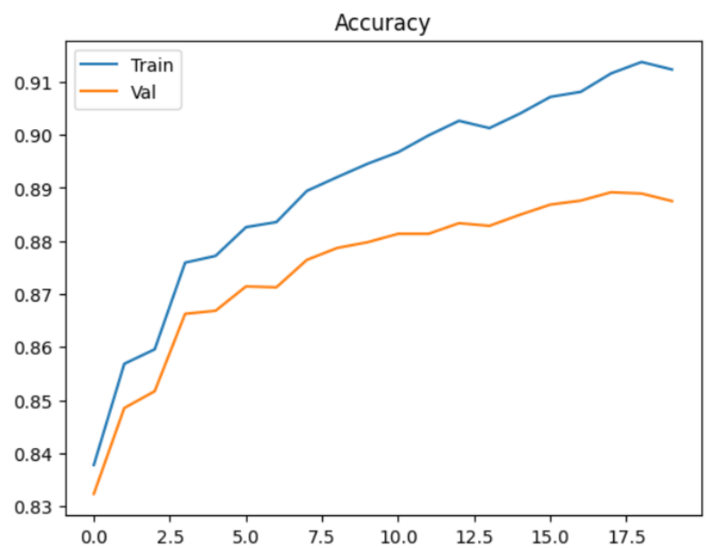
Sequential batch train:

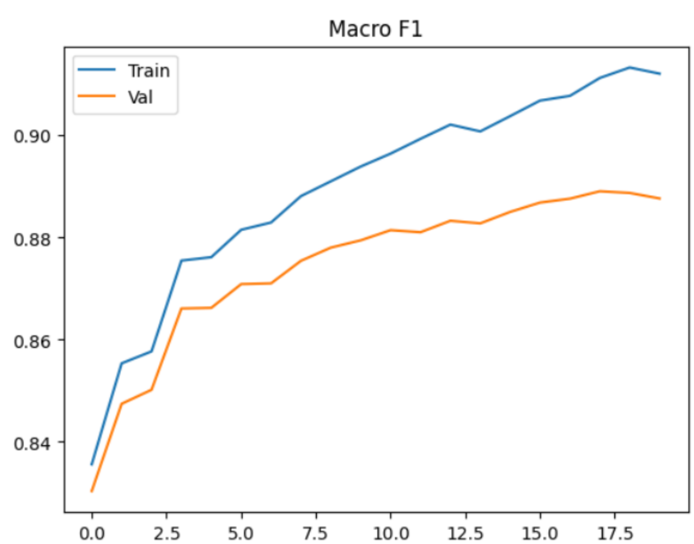
```
Model 3. 3
alpha:  0.00078125 decay: 0.9999 dropout: 0.5 optimizer: ADAM  epoch: 20
```

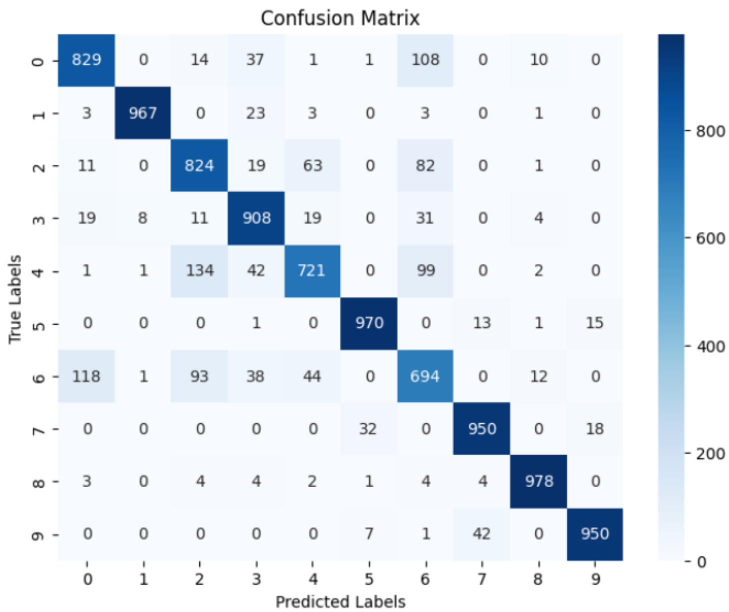
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.912291	0.234755	0.912083
Validation	0.8875	0.313280	0.887647
Test	0.8791	0.337976	0.878953


Loss
Train
Val
0.45
0.40
0.35
0.30
0.25
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Accuracy
Train
Val
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.84
0.83
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Macro F1
Train
Val
0.90
0.88
0.86
0.84
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

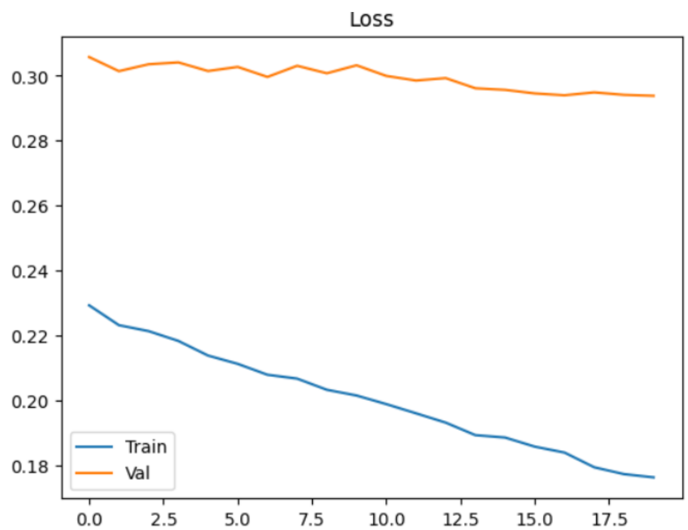


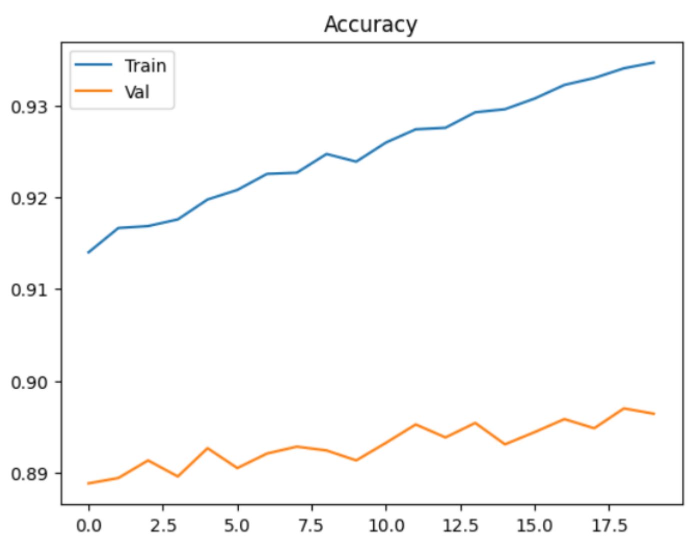
Random batch train:

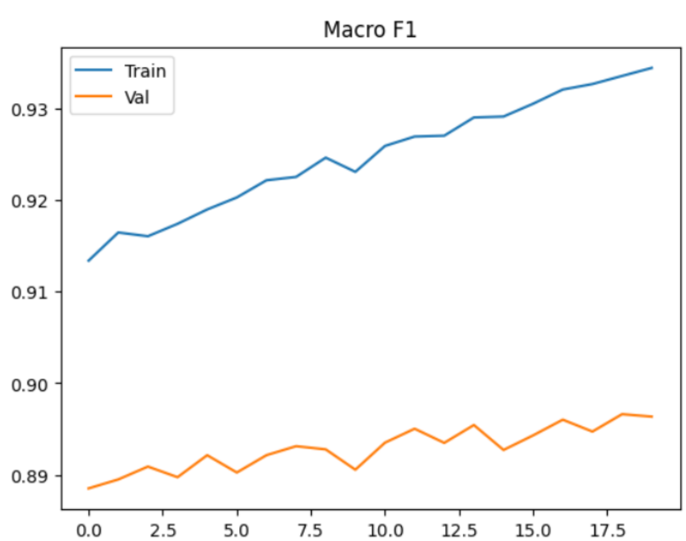
```
Model 3. 3
alpha:  0.00078125 decay: 0.9999 dropout: 0.5 optimizer: ADAM  epoch: 20
```

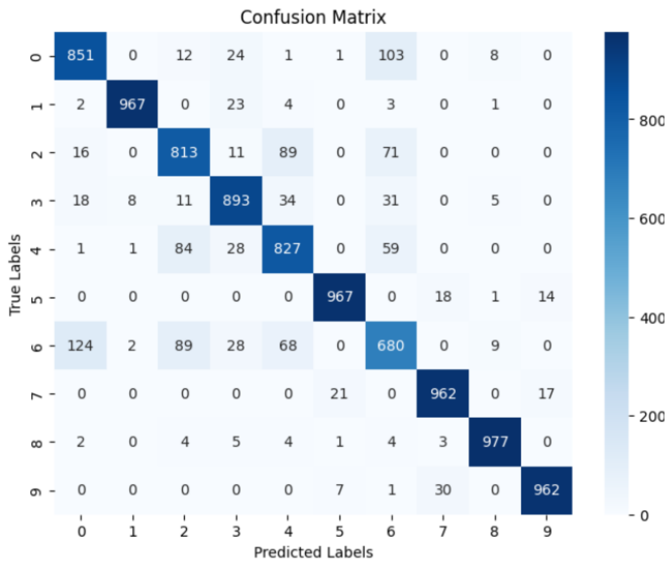
Performance

	Accuracy	Loss	Macro-F1-score
Train	0.9346666	0.176164	0.934438
Validation	0.896416	0.293814	0.896344
Test	0.8899	0.317037	0.889709


Loss
Train
Val
0.30
0.28
0.26
0.24
0.22
0.20
0.18
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Accuracy
Train
Val
0.93
0.92
0.91
0.90
0.89
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Macro F1
Train
Val
0.93
0.92
0.91
0.90
0.89
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5



Best Model:

Model 2

Alpha: 0.00078125 (0.1/128)

Decay: 0.9999

Dropout: 0.2

Optimizer: YOGI

Batch Size: 128

Random Train with Epoch: 10

Performance

	Accuracy	**Loss**	**Macro-F1-score**
Validation	0.90025	0.290522	0.899809
Test	0.8925	0.318876	0.891873

Random Train with Epoch: 15

Performance

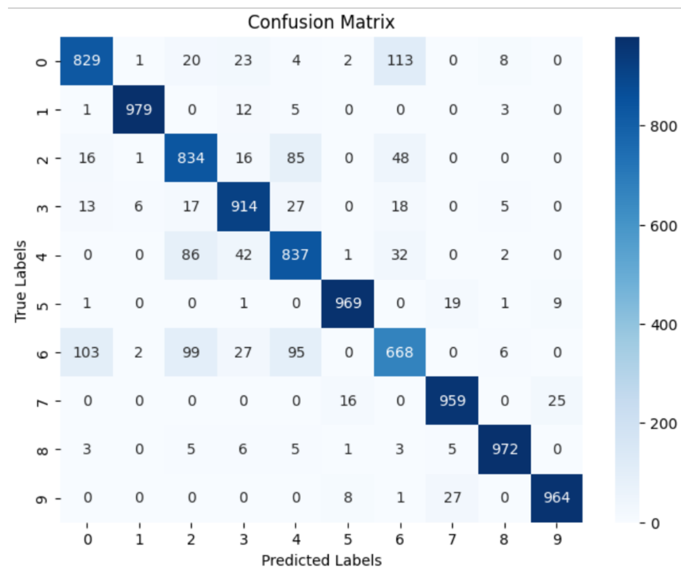
	Accuracy	**Loss**	**Macro-F1-score**
Validation	0.90175	0.284786	0.902258

Random Train with Epoch: 30

Performance

	Accuracy	Loss	Macro-F1-score
Validation	0.90475	0.292950	0.904722
Test	0.8985	0.315321	0.898469

Confusion matrix (epoch 10):



Confusion matrix (epoch 30):